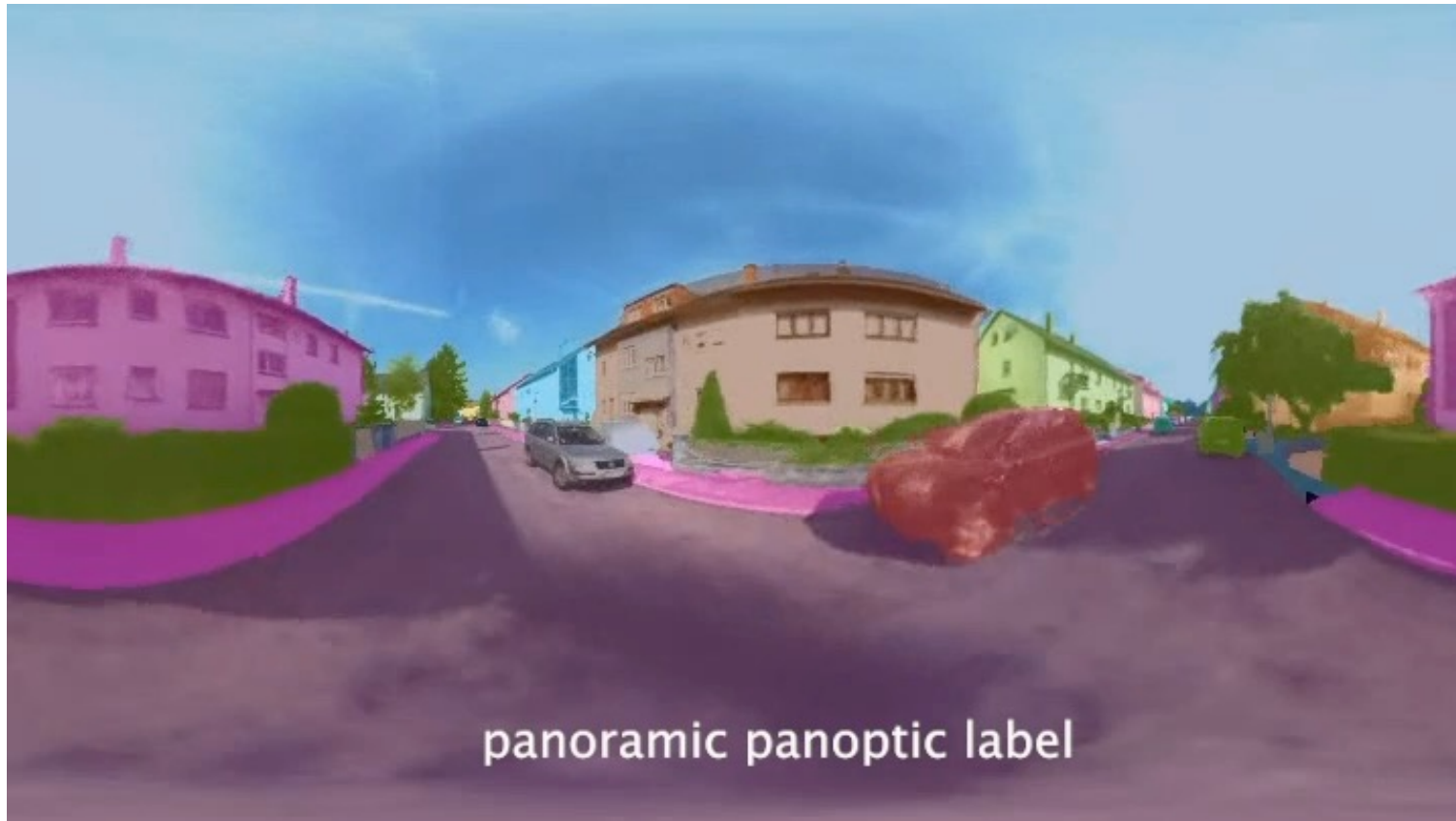




Panoramic Label Rendering for Autonomous Driving Perception and Simulation

Xiao FU

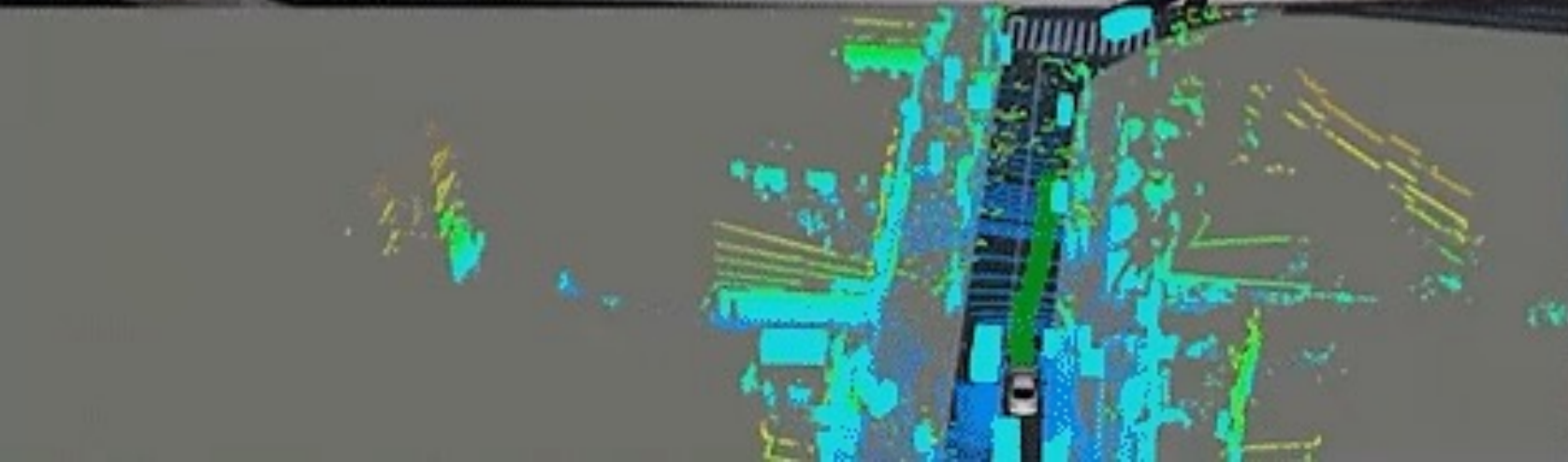
Oct 20, 2023 @ Shenlan Open Courses



15:19:22:15

3X SPEED

滴滴自动驾驶
DIDI AUTONOMOUS DRIVING



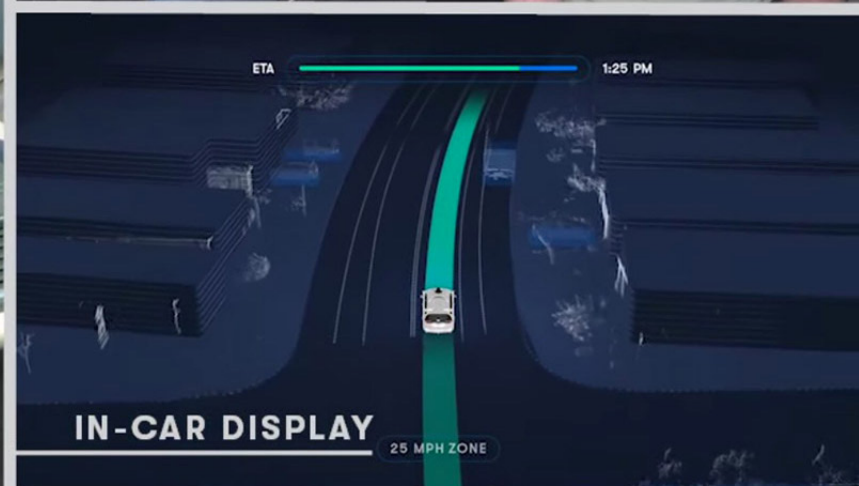




WAYMO 03



RIDERS



IN-CAR DISPLAY



FRONT VIEW



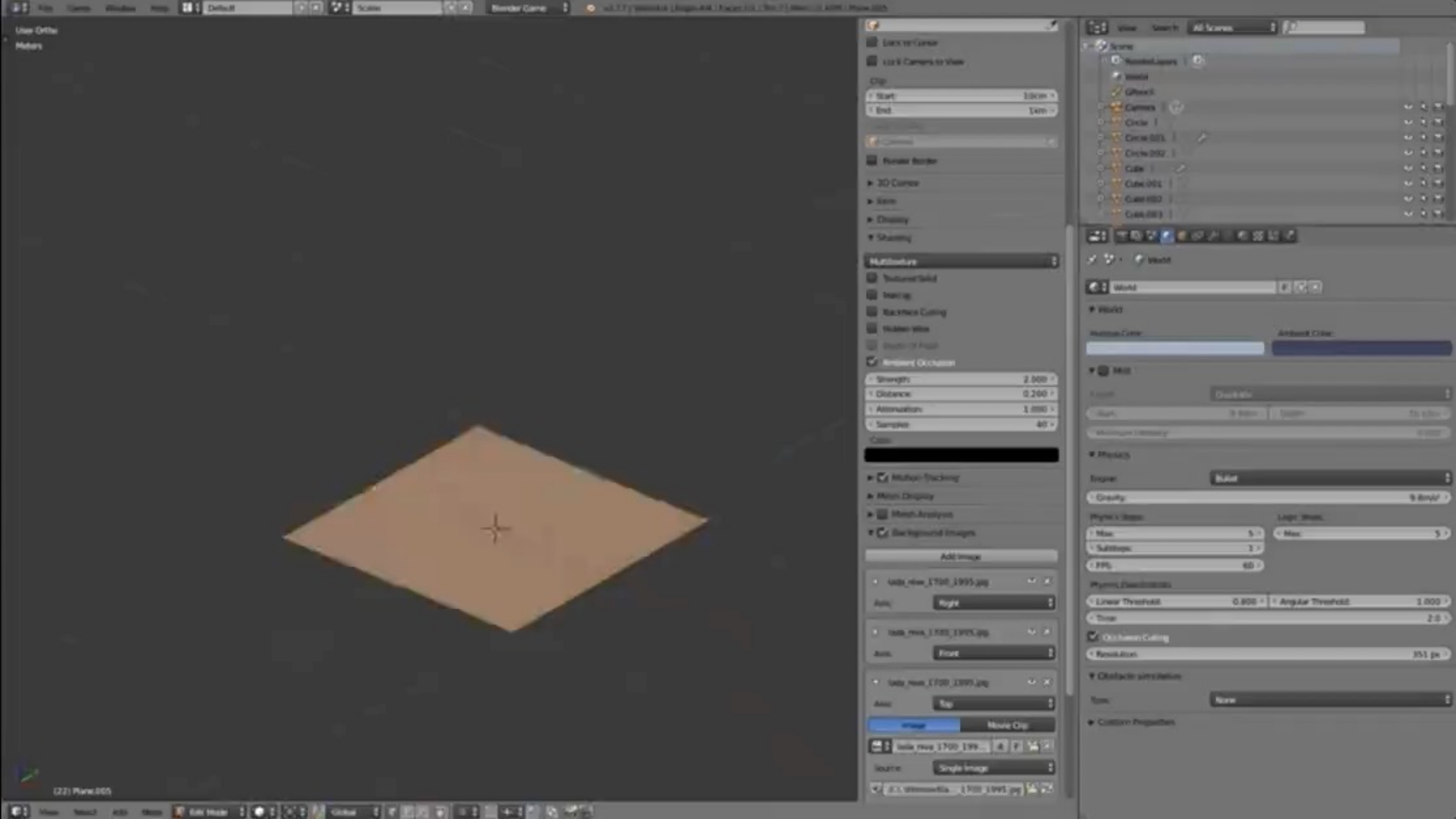




51Sim is committed to providing



CARLA
Release 0.9.13

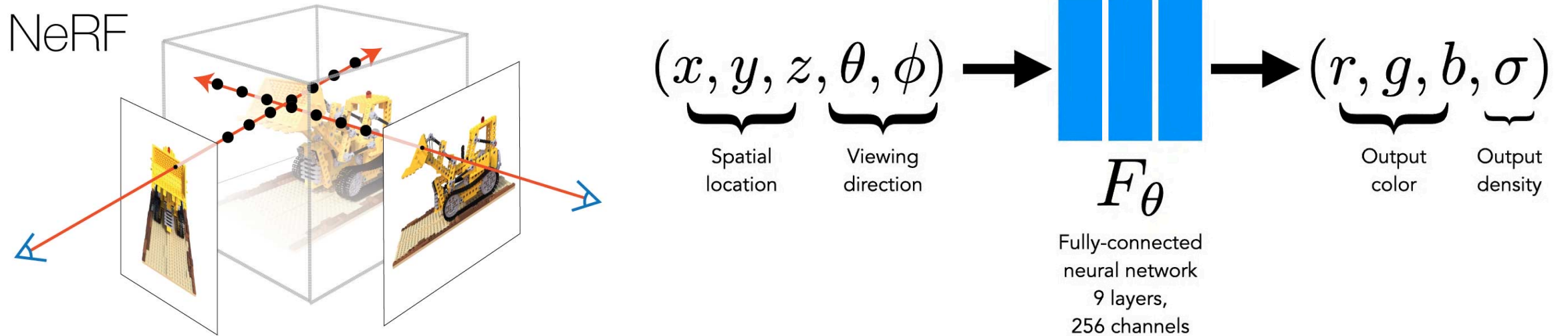


Neural Radiance Fields



- Neural Radiance Fields (NeRF) achieve **high-fidelity** novel view synthesis
- Can we build a simulator from the **real world**?

Neural Radiance Fields



- Neural Radiance Fields (NeRF) achieve **high-fidelity** novel view synthesis

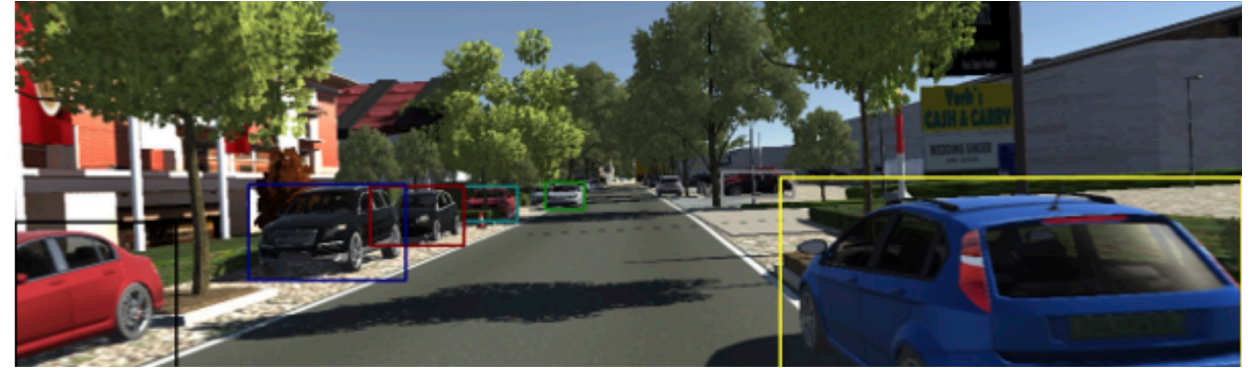
Autonomous Driving Simulator

Manual Design:

- + 3D controllable, perfect label
- Domain gap, expensive to create

Neural Rendering:

- + No domain gap, no design effort
- Absence of semantic
- Lack of high-fidelity reconstruction

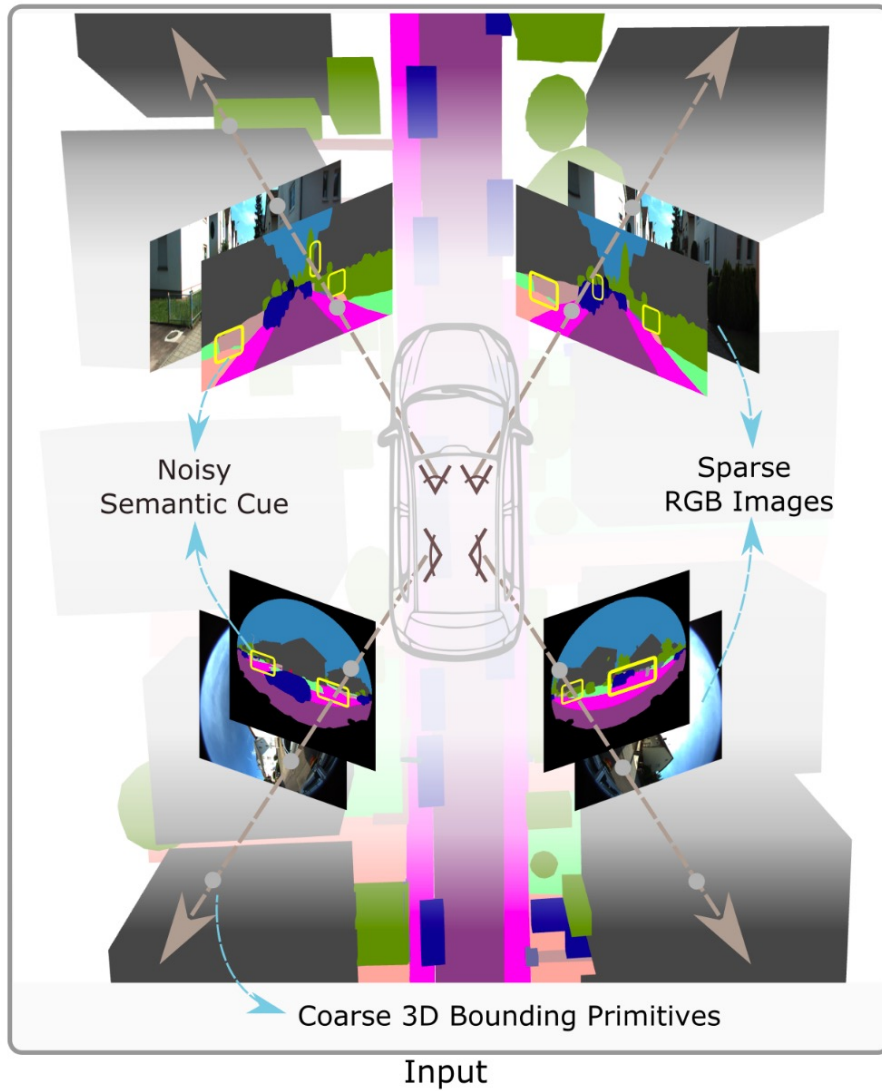


Virtual KITTI [Gaidon et al.]

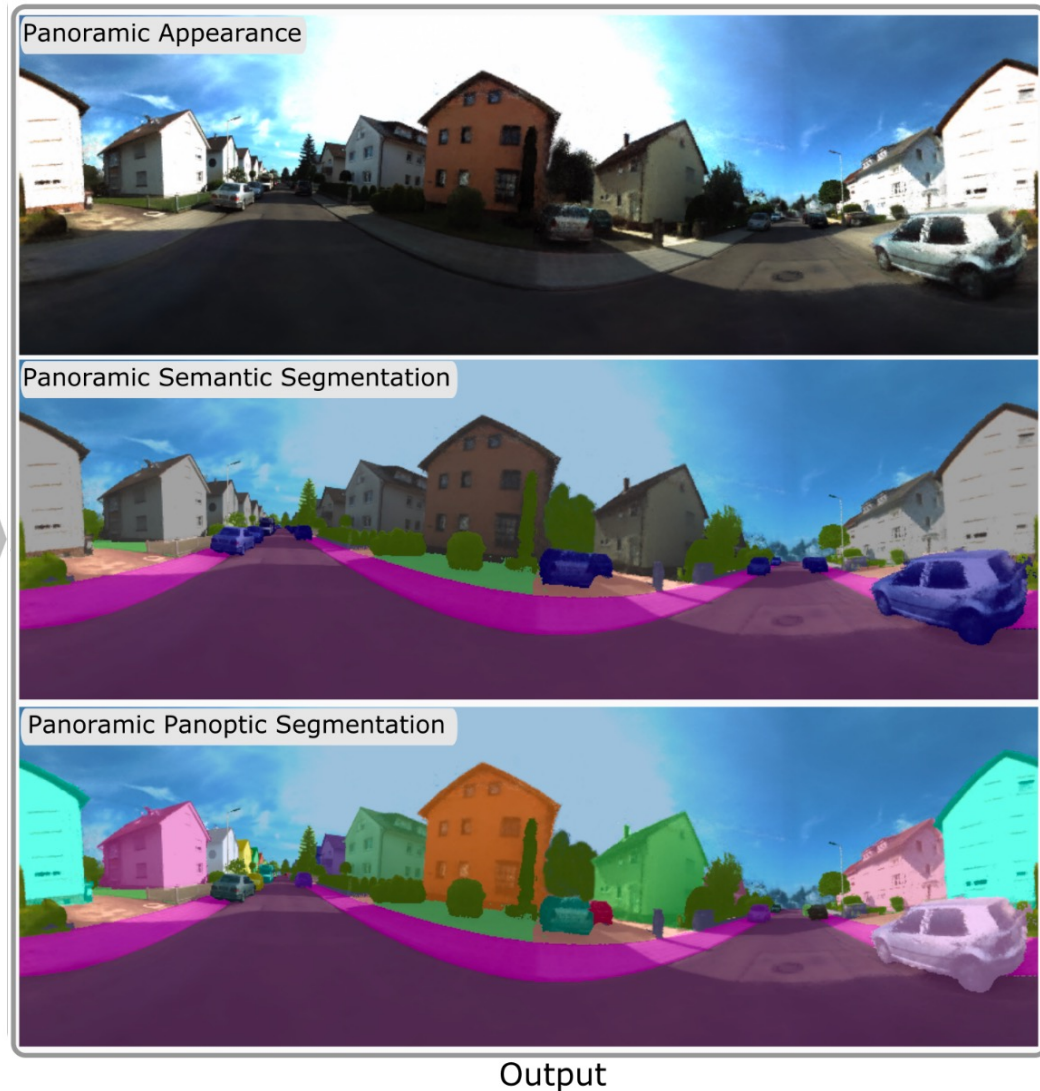


KITTI [Geiger et al.]

PanopticNeRF-360



PanopticNeRF-360



- Idea: Infer semantic & instance label in 3D space, obtain **omnidirectional** 2D panoptic labels via **volume rendering**
- Input: Sparse perspective & fisheye images, coarse 3D bounding primitives, noisy 2D semantic labels

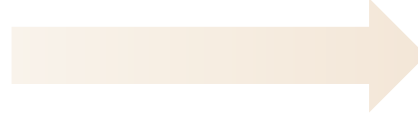
Motivation



Images



Per-frame
2D Annotation



Panoptic Label



1.5h / frame



Expensive

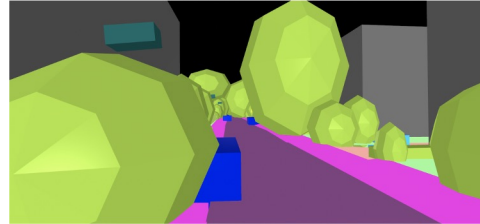


Labor-intensive

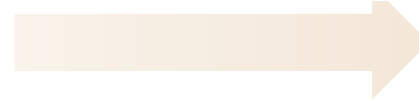
Motivation



Images



Coarse 3D
Annotation



Panoptic Label



0.75min / frame



Cheap

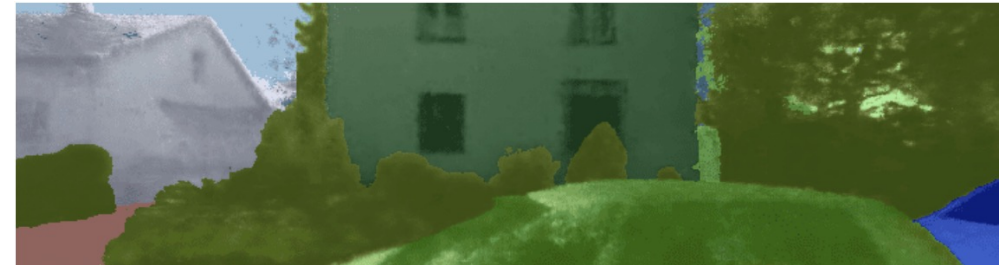
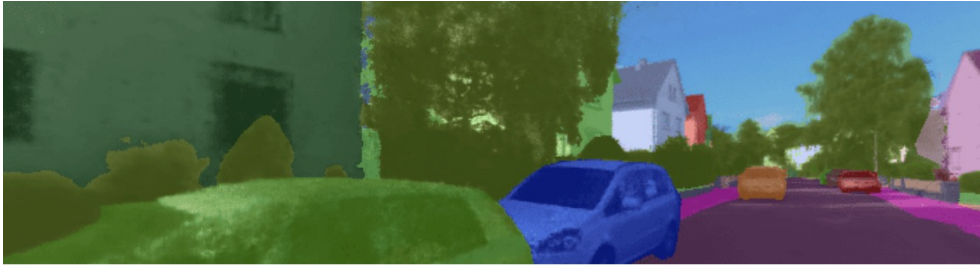


Efficient

Motivation



Pre-recorded Viewpoint



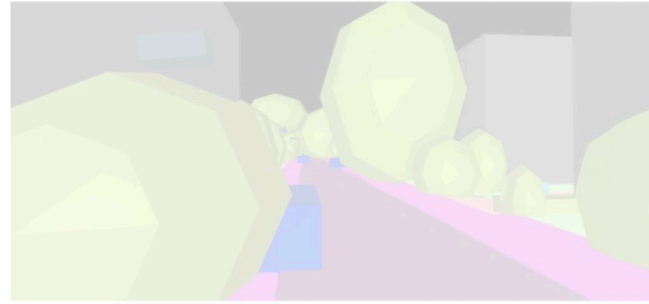
Novel Viewpoints

Offering RGB images and annotations from **arbitrary novel viewpoints**

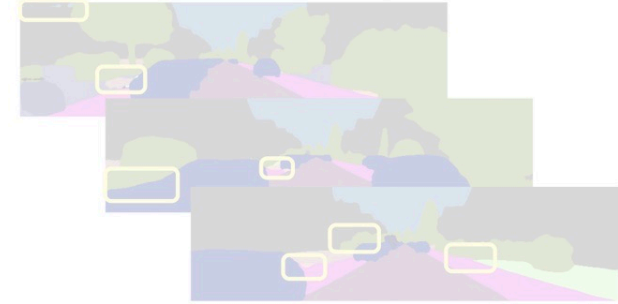
Motivation



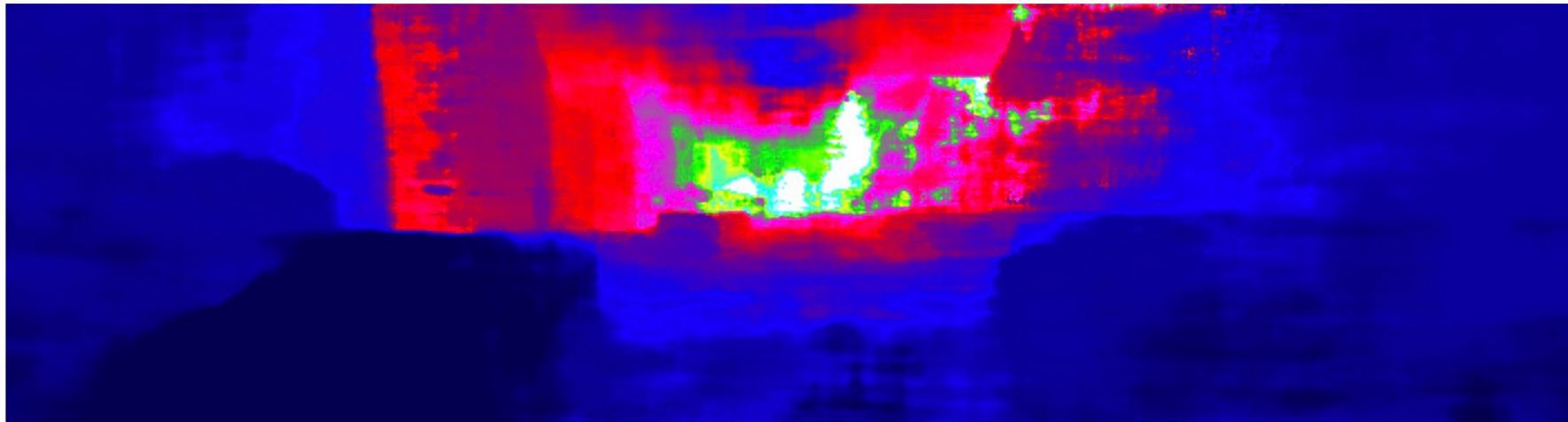
Sparse Images



Coarse 3D Bounding Primitives



Noisy 2D Predictions



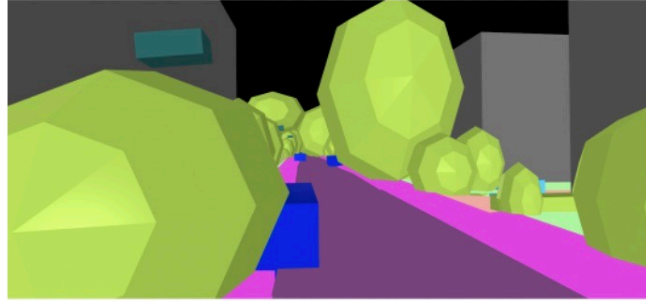
NeRF [Mildenhall et al., ECCV 2020]

NeRF fails to recover good geometry given sparse views

Motivation



Sparse Images



Coarse 3D Bounding Primitives



Noisy 2D Predictions



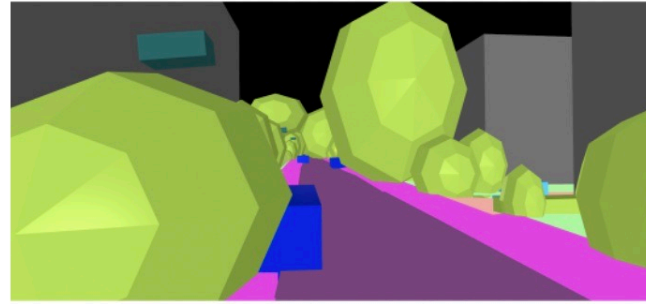
NeRF* [Mildenhall et al., ECCV 2020]

NeRF fails to recover good geometry given sparse views

Motivation



Sparse Images



Coarse 3D Bounding Primitives



Noisy 2D Predictions



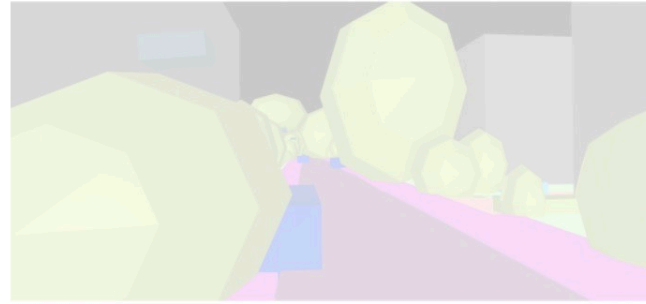
NeRF* [Mildenhall et al., ECCV 2020]

3D Label is ambiguous at intersections of bounding primitives

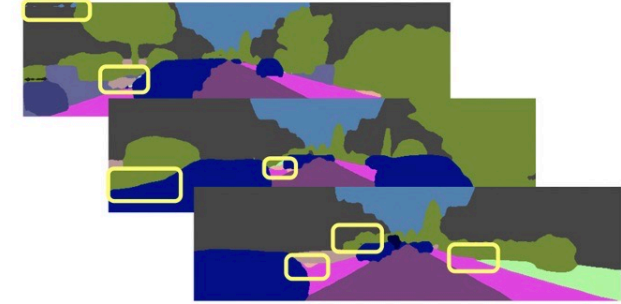
Challenge



Sparse Images



Coarse 3D Bounding Primitives



Noisy 2D Predictions



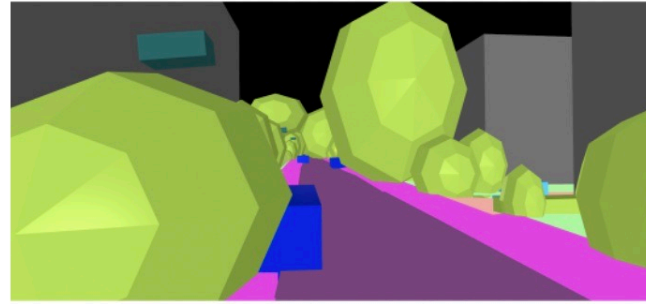
Semantic NeRF [Zhi et al., ICCV 2021]

Denoising is hard given bad geometry & predictions of pre-trained 2D models

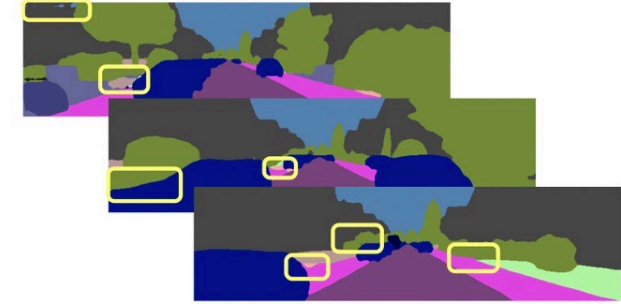
Challenge



Sparse Images



Coarse 3D Bounding Primitives



Noisy 2D Predictions



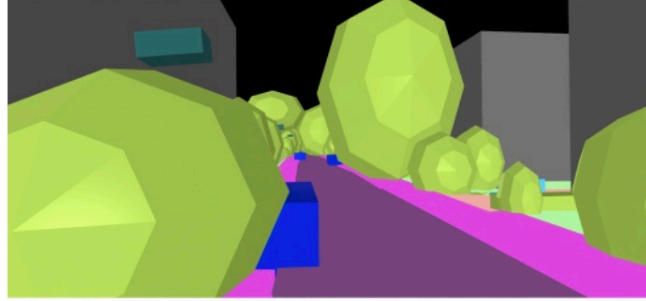
[Liao et al., TPAMI 2022]

Existing 3D-to-2D label transfer methods suffer from inaccurate reconstruction

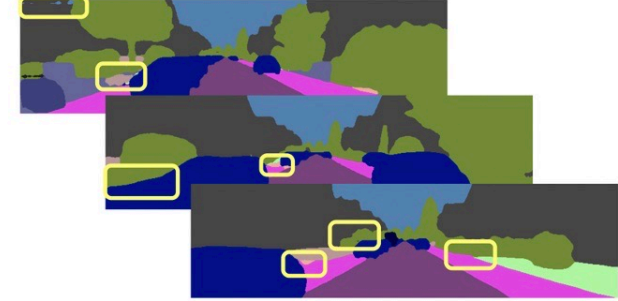
Challenge



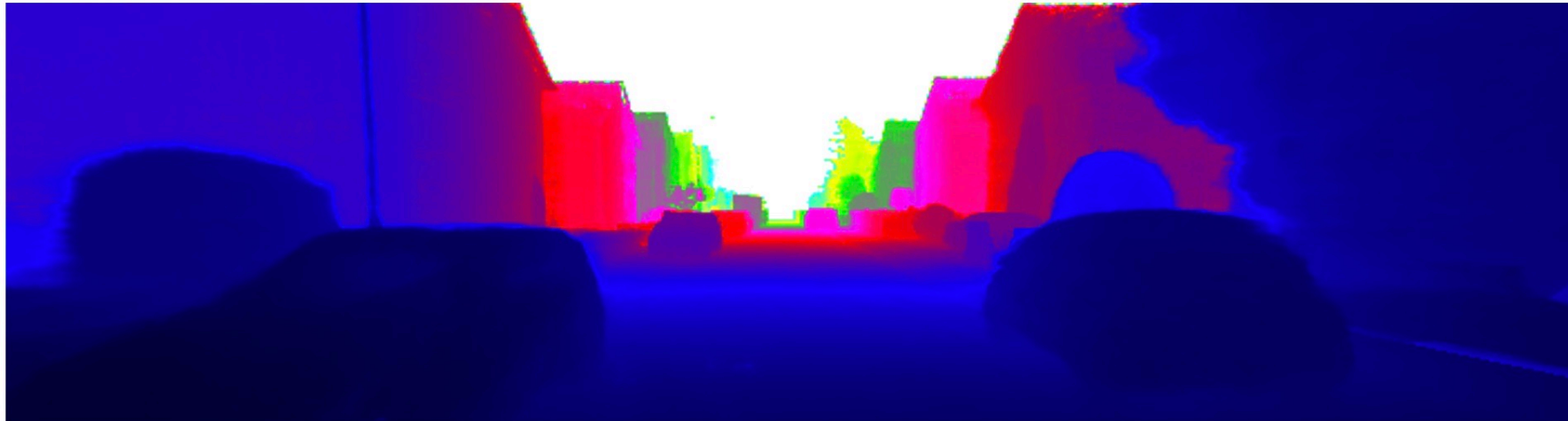
Sparse Images



Coarse 3D Bounding Primitives



Noisy 2D Predictions



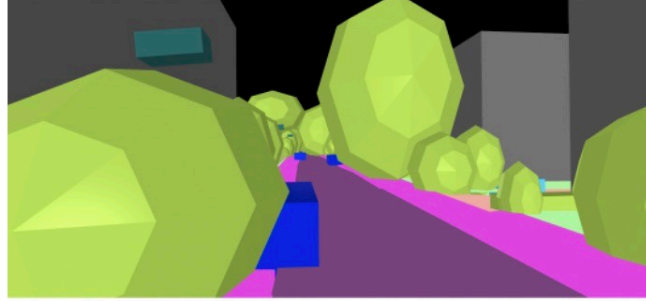
Ours

PanopticNeRF-360 improves geometry guided by semantic information ...

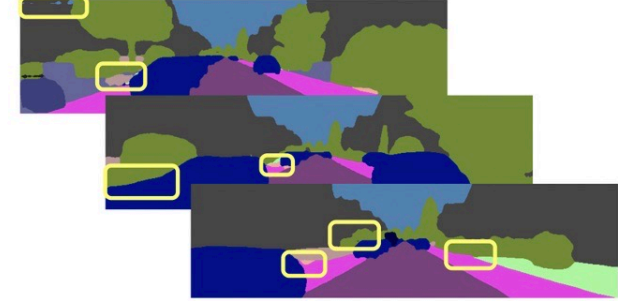
Challenge



Sparse Images



Coarse 3D Bounding Primitives



Noisy 2D Predictions



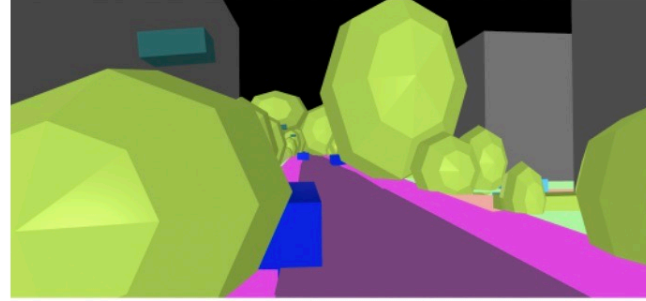
Ours

.. and resolves label ambiguity by unifying 3D and 2D semantic information

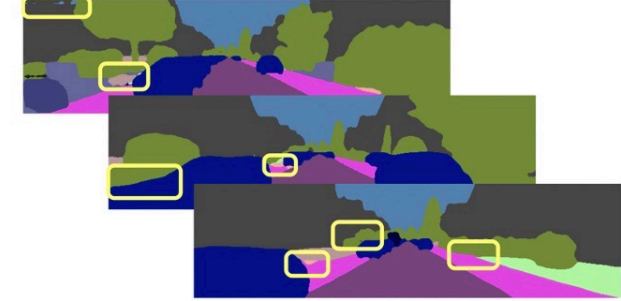
Challenge



Sparse Images



Coarse 3D Bounding Primitives



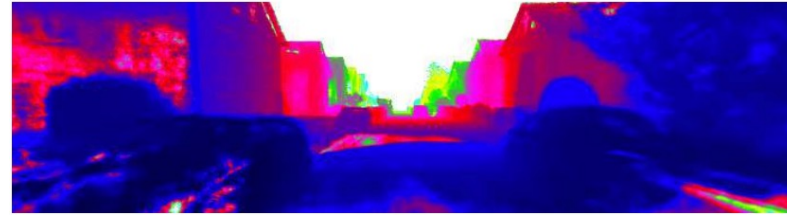
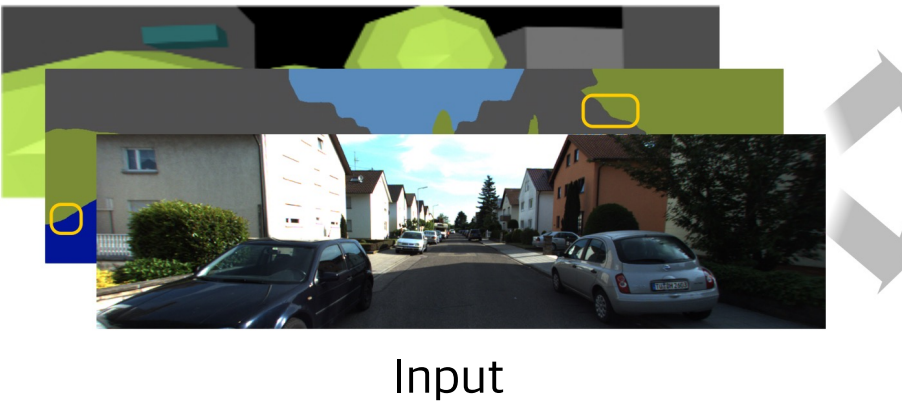
Noisy 2D Predictions



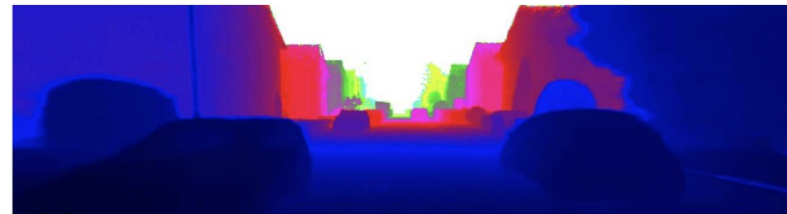
Ours

.. and resolves label ambiguity by unifying 3D and 2D semantic information

Challenge



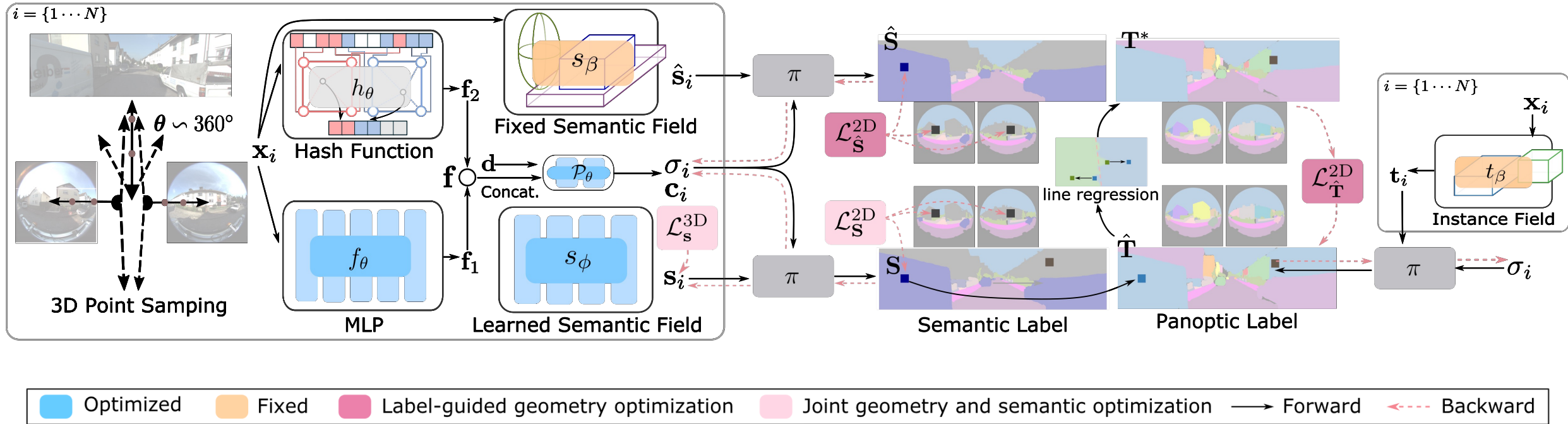
Naïve Joint Optimizaion of Geometry and Semantics



Label-Guided Mutual-Enhanced Optimizaion of Geometry and Semantics

Key idea : mutual enhancement of geometry and semantics
by exploiting the complementarity of 3D and 2D priors

Method

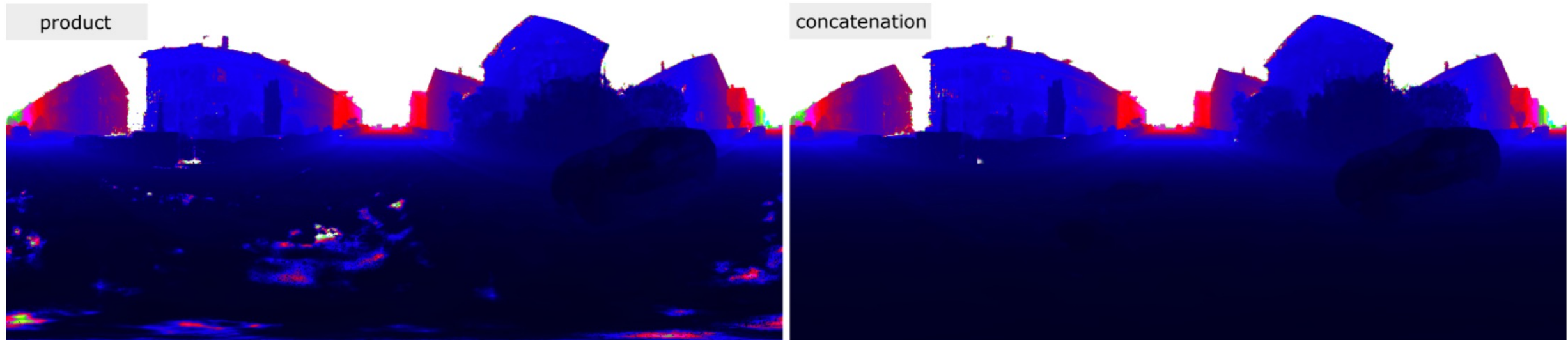


- Sample in 3D space within **coarse 3D bounding boxes**
- Label-guided **geometry optimization** based on **fixed semantic & instance fields**
- Joint semantic and geometry optimization based on **learned semantic field** to resolve label ambiguity

Hybrid Scene Representation

	semantics	appearance	convergence
► MLPs :	high (inductive smoothness bias)	low	low
► Hash Grids :	medium	high (multi-res. local details)	high

How to strike a balance between semantics and appearance ?



Hybrid Scene Representation



Ours (panoptic label & appearance)



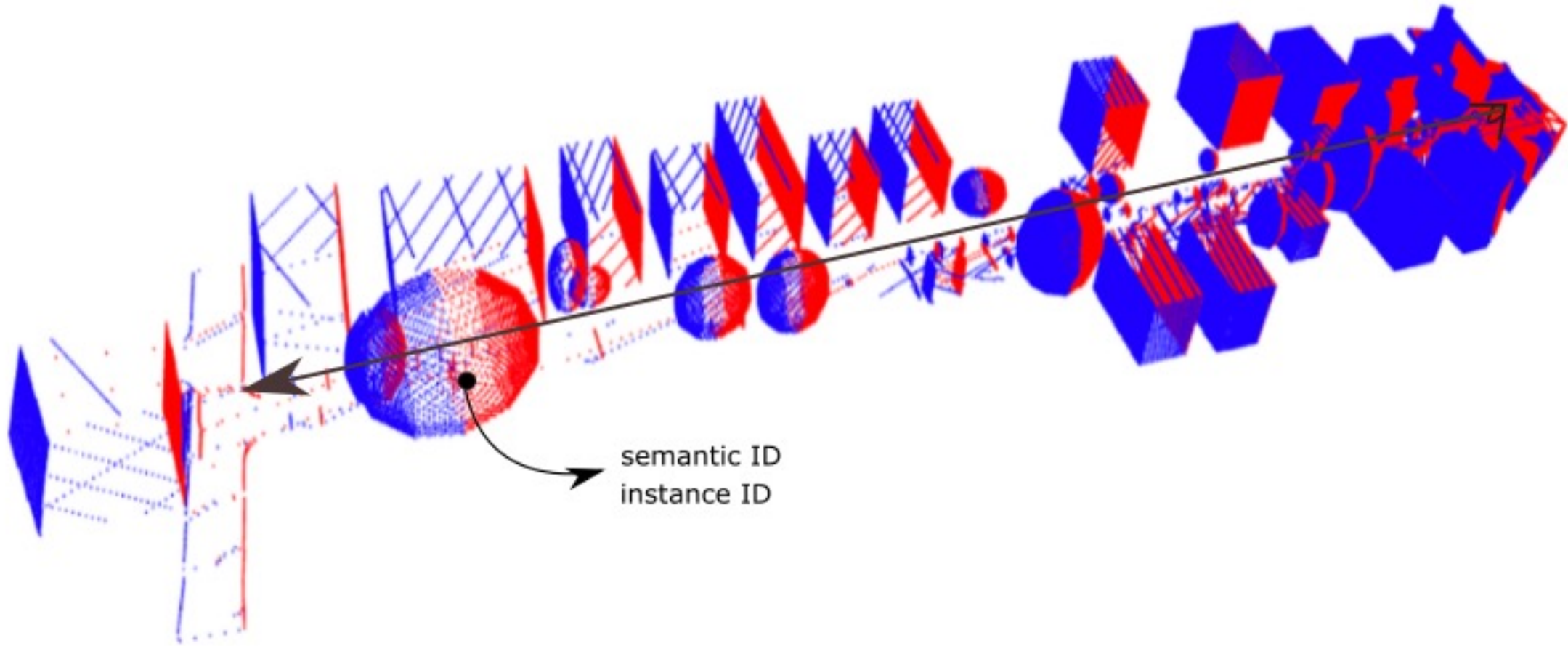
MLP (panoptic label & appearance)



Triplane (panoptic label & appearance)

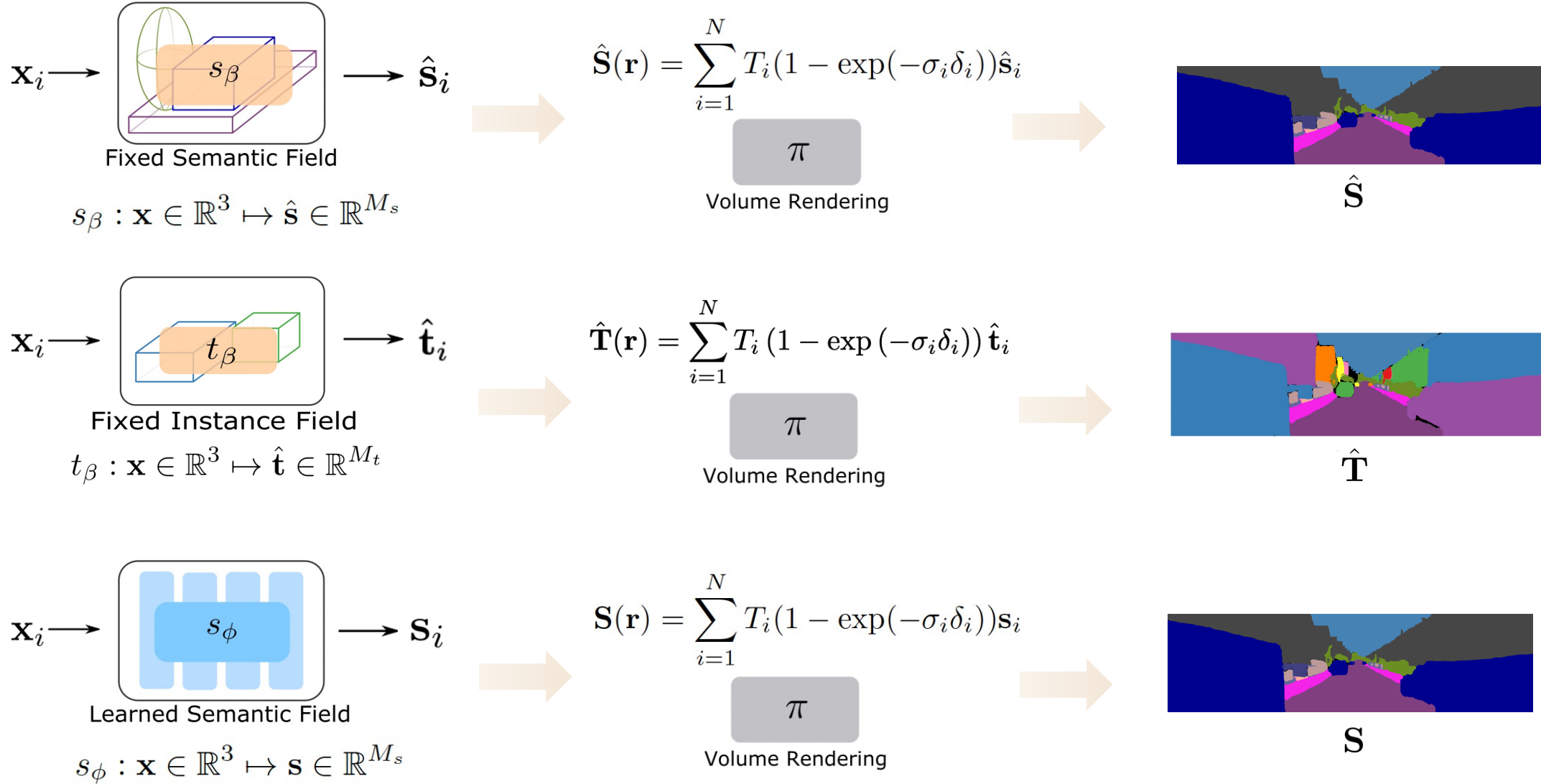


Dual Semantic Fields & Fixed Instance Field

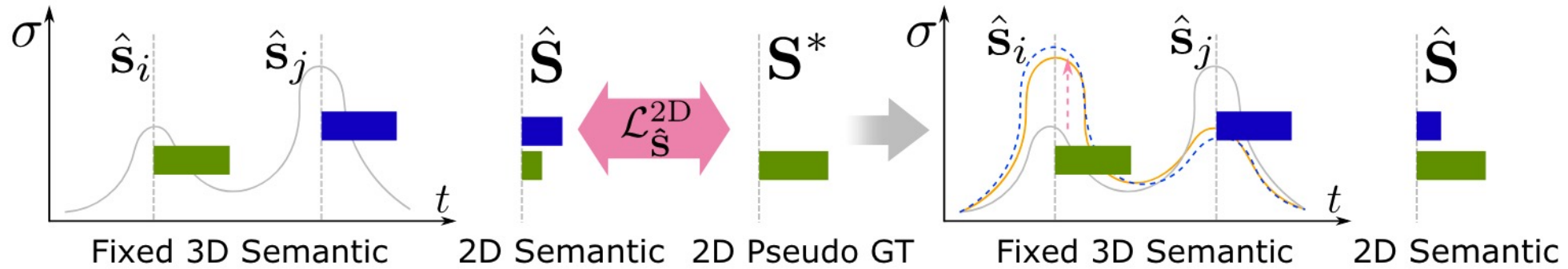


Ray-primitive intersections : The red dots and blue dots indicate where the rays hit into and out of the coarse primitives, respectively.

Dual Semantic Fields & Fixed Instance Field

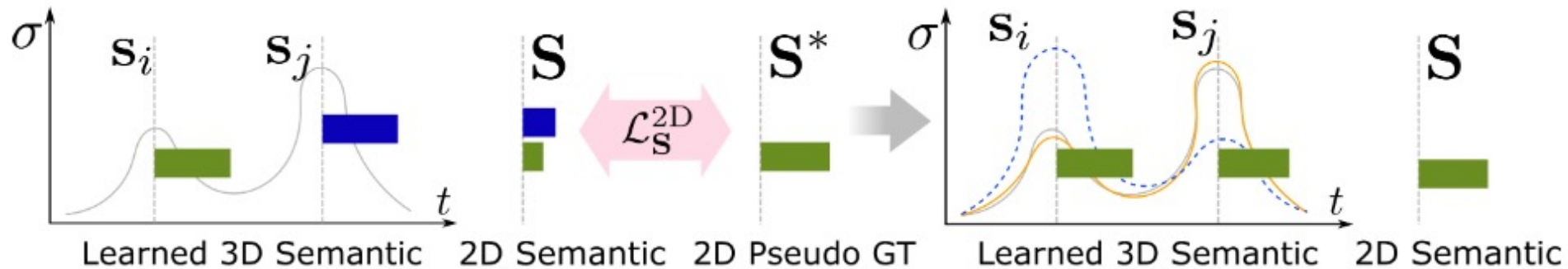


Panoptic Label-Guided Geometry Optimization



- A single ray of fixed semantic field. Loss can only update the density as the semantics logits are fixed.

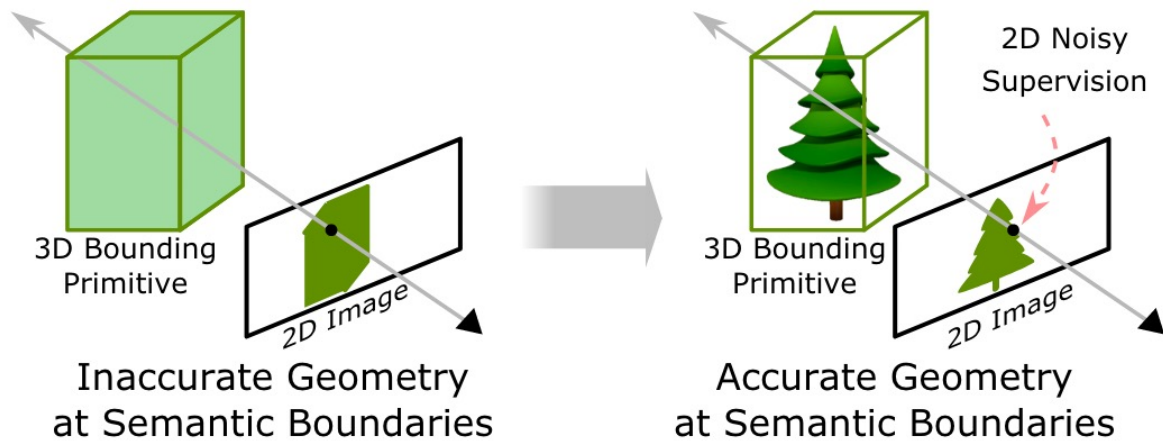
$$\mathcal{L}_{\hat{\mathbf{S}}}^{2D}(\theta, \beta) = -\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} \sum_{k=1}^{M_s} \mathbf{S}_k^*(\mathbf{r}) \log \hat{\mathbf{S}}_k(\mathbf{r})$$



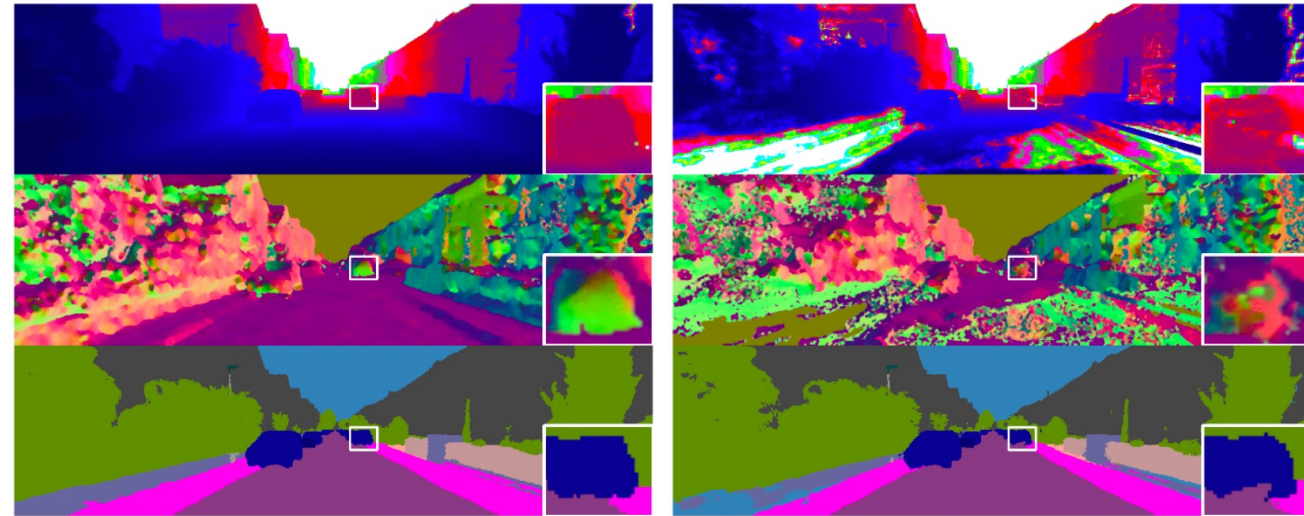
- A single ray of learned semantic field. Loss can both update the density and the semantics.

$$\mathcal{L}_{\mathbf{S}}^{2D}(\theta, \phi) = -\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} \mathbb{1}(\mathbf{r}) \sum_{k=1}^{M_s} w(k) \mathbf{S}_k^*(\mathbf{r}) \log \mathbf{S}_k(\mathbf{r})$$

Panoptic Label-Guided Geometry Optimization



Input

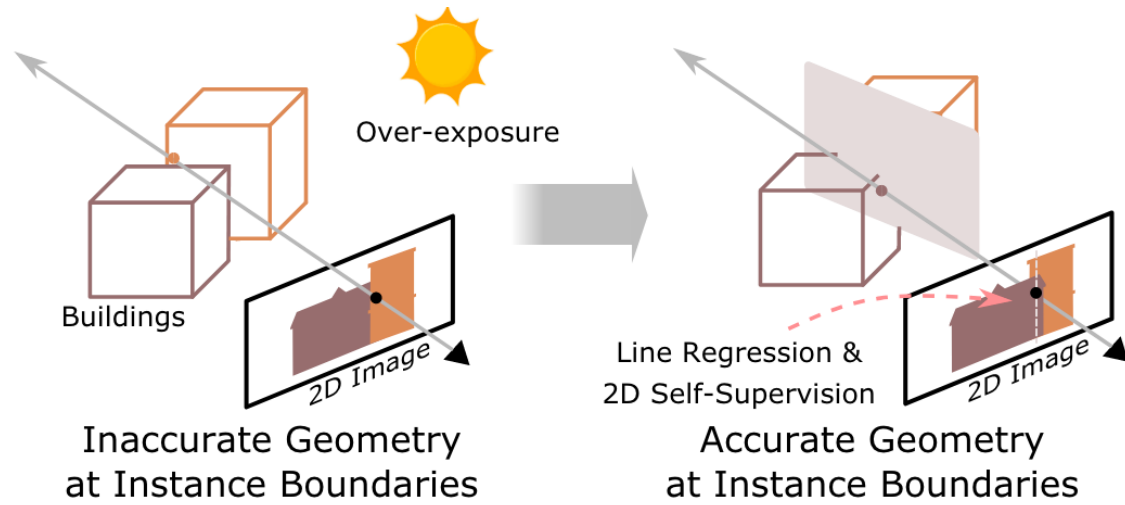


Complete

w/o $\mathcal{L}_{\hat{S}}^{2D}$

► Semantic-guided geometry optimization

Panoptic Label-Guided Geometry Optimization



- 1) “building” is the most frequent connected thing class that may yield wrong geometry
- 2) cuboid-like “building” has line-shaped boundaries



(a) Complete model (b) w/o finetuning instance

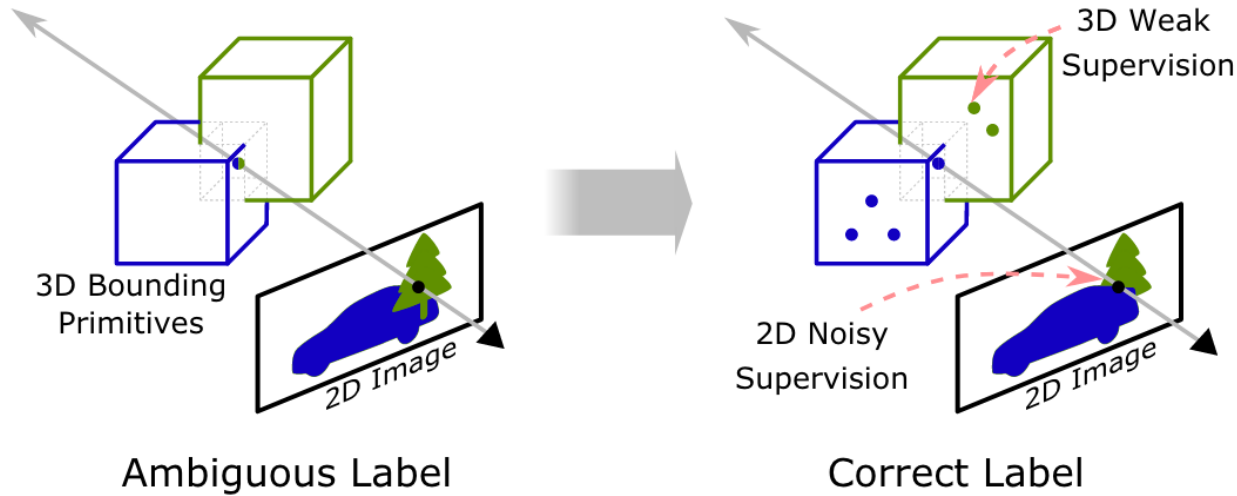
► Instance-guided geometry optimization

Panoptic Label-Guided Geometry Optimization



➤ Instance-guided geometry optimization

Geometry-Guided Joint 3D and 2D Semantic Optimization



Overlapped regions : 2D Noisy Supervision

$$\mathcal{L}_S^{2D}(\theta, \phi) = -\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} \mathbb{1}(\mathbf{r}) \sum_{k=1}^{M_s} w(k) \mathbf{S}_k^*(\mathbf{r}) \log \mathbf{S}_k(\mathbf{r})$$

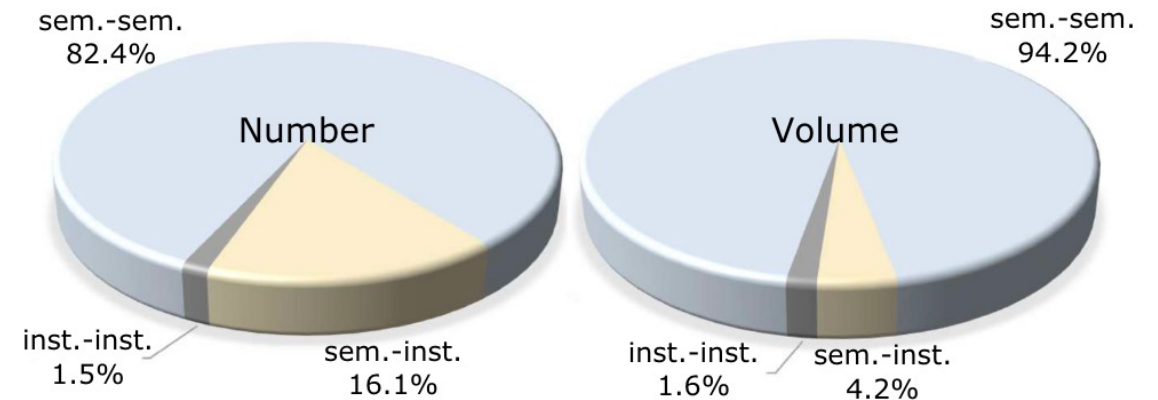
Resolve label ambiguity

Non-overlapped regions : 3D Weak Supervision

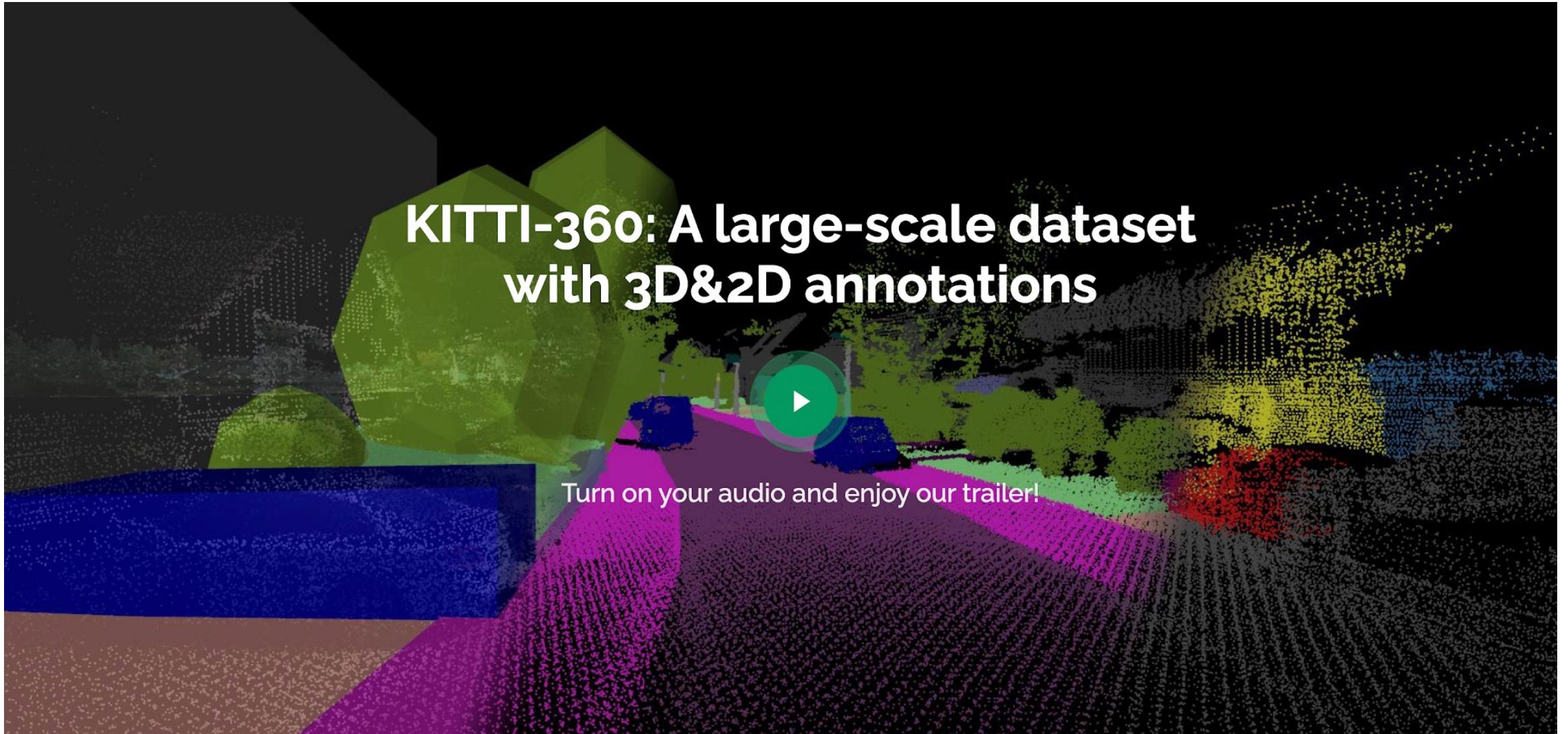
$$\mathcal{L}_s^{3D}(\theta, \phi, \beta) = -\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} \sum_{i=1}^N \mathbb{1}(\mathbf{x}) \sum_{k=1}^{M_s} \hat{\mathbf{s}}_i^k \log \mathbf{s}_i^k$$

Increase stability of semantic prediction

Why not perform learned instance optimization ?



The overlapped instance regions are small



KITTI-360 dataset : the only available dataset **fully covering 3D bounding primitives** in urban scenes.

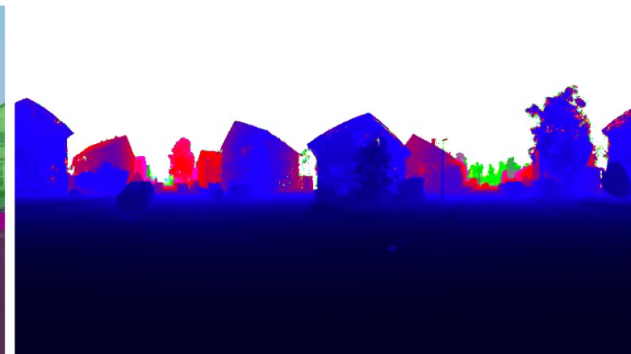
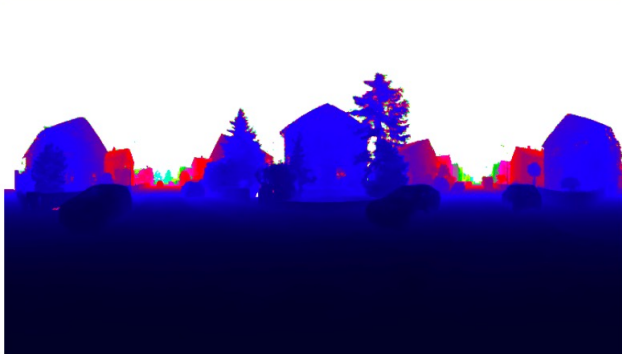
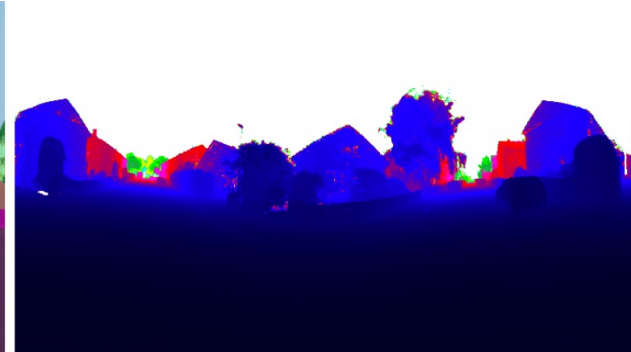
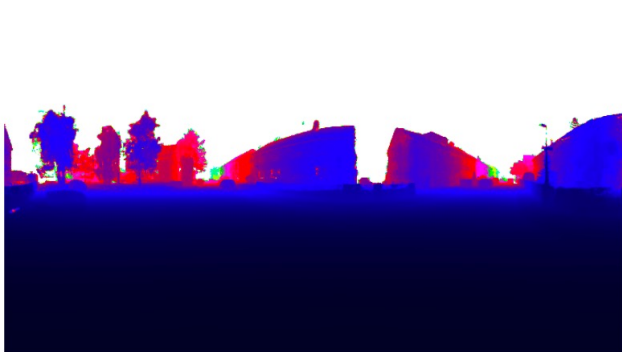
Panoptic Label Rendering on Perspective Views



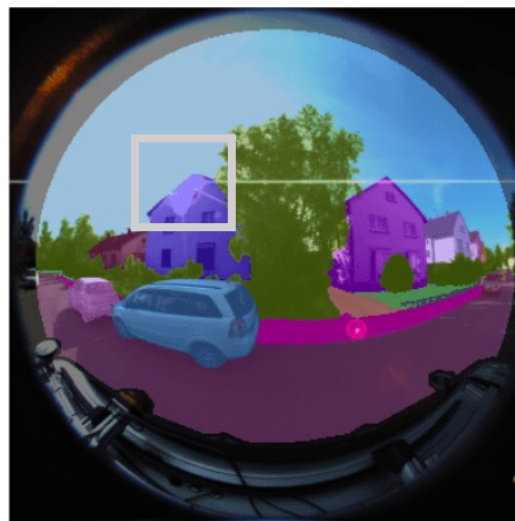
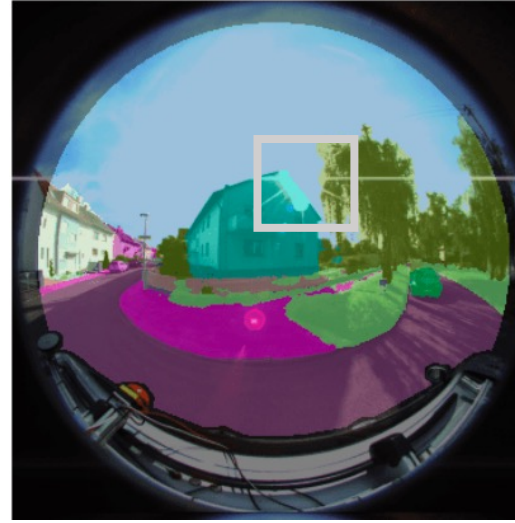
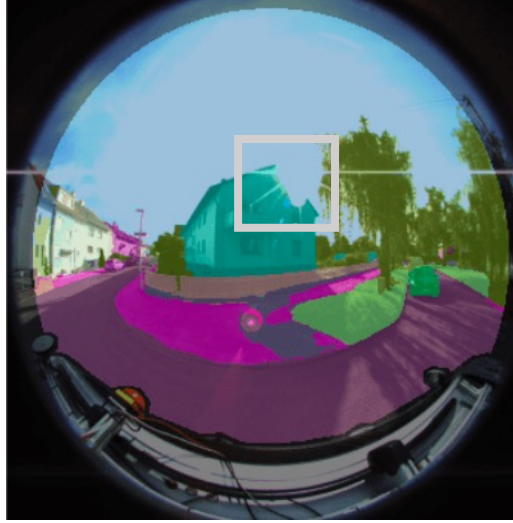
Panoptic Label Rendering on Fisheye Views



Panoramic Panoptic Label & Depth Rendering



Label Rendering at Overexposed Regions (Missing Parts)



Input

3D-to-2D CRF

Ours

Self-distilled Pseudo GT Using SAM



Top : input RGB **Middle** : fixed instance map **Bottom** : SAM results using “Middle” as input

SAM results perform worse than line-shaped assumption to generate Pseudo-panoptic labels

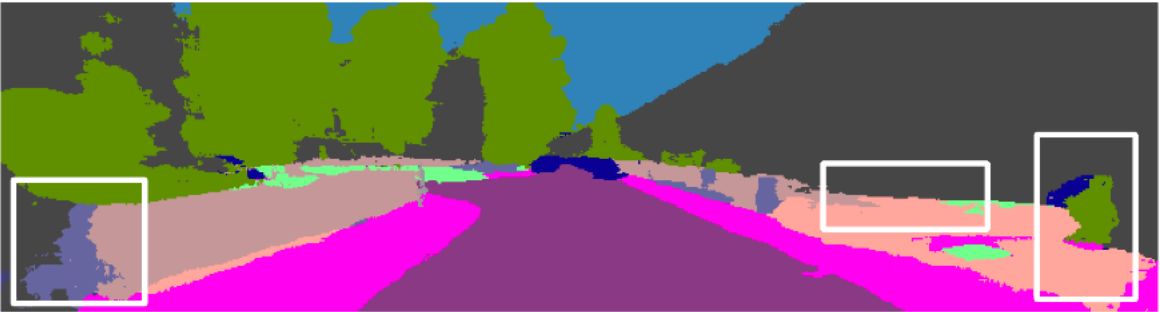
Semantic Comparison on Perspective Views



Input



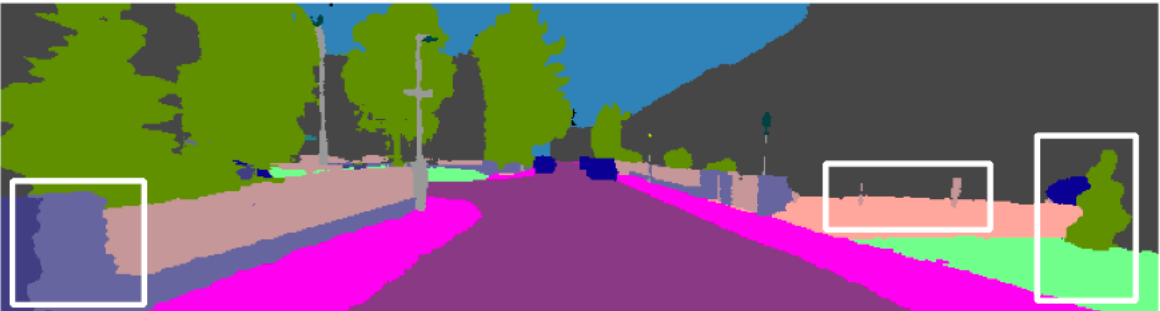
PSPNet



JacobiNeRF



3D-to-2D CRF

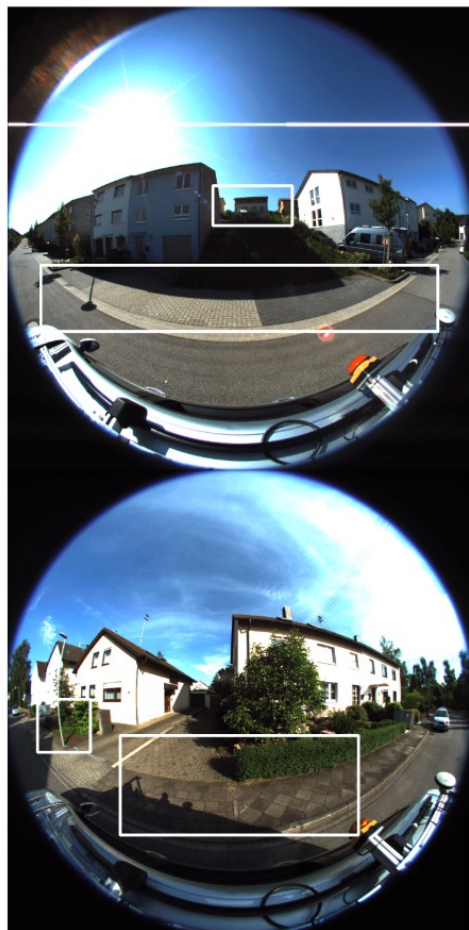


Ours

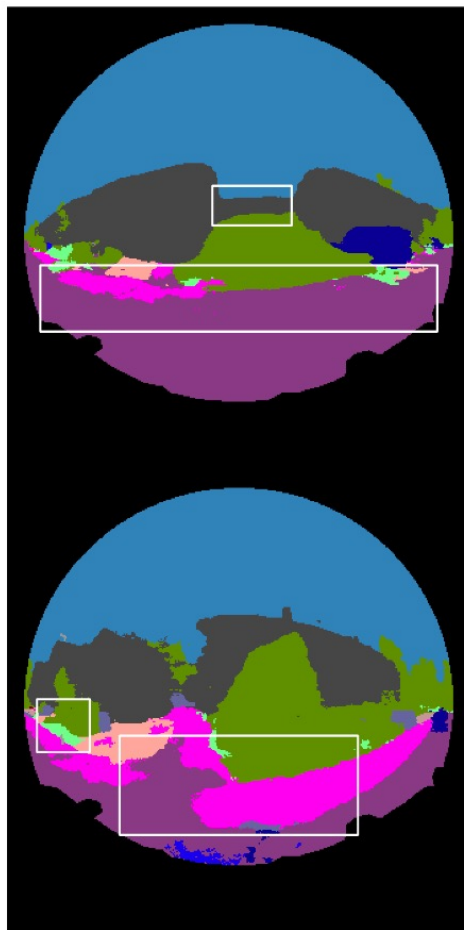


GT

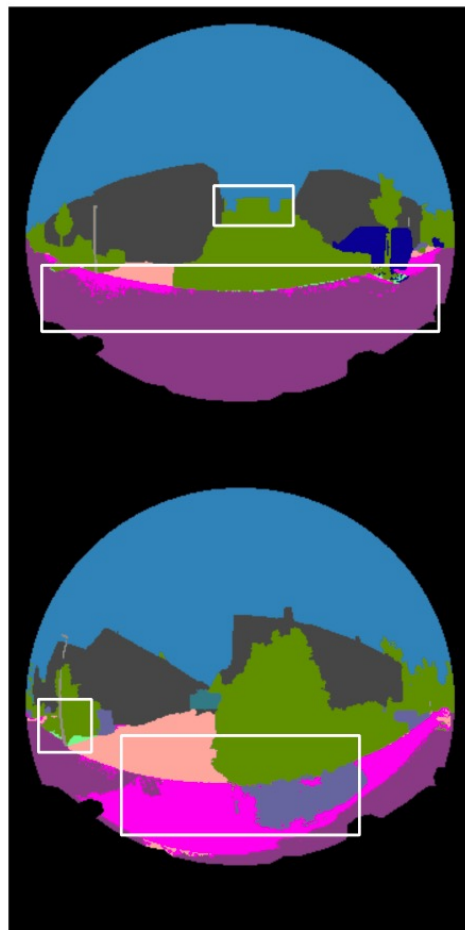
Semantic Comparison on Fisheye Views



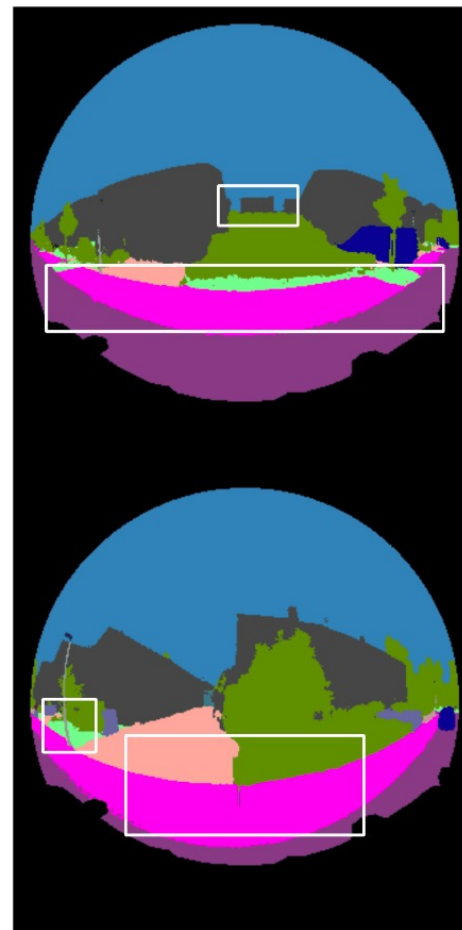
Input



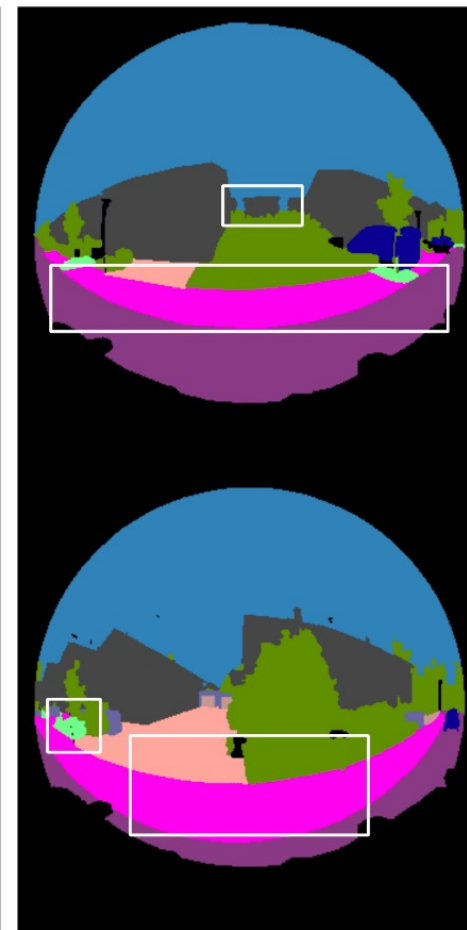
JacobiNeRF



3D-to-2D CRF



Ours

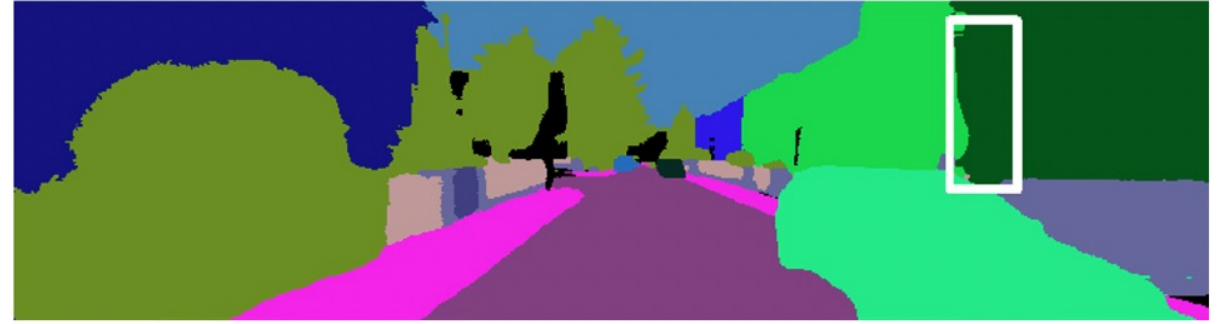


GT

Panoptic Comparison on Perspective Views



Input



3D-2D CRF



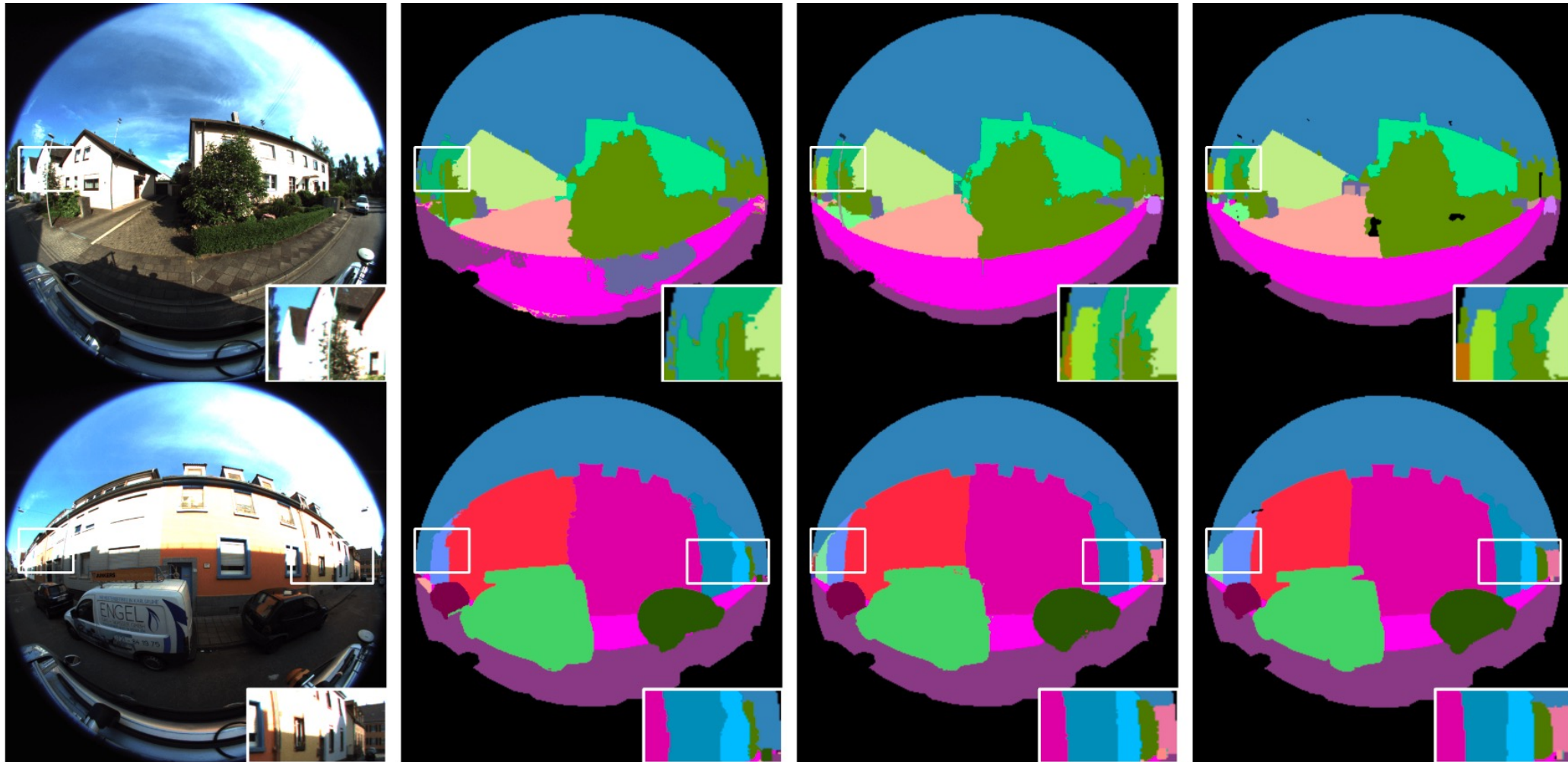
Ours



GT

► Inferring in the 2D image space suffers from **overexposure**

Panoptic Comparison on Fisheye Views



Input

3D-to-2D CRF

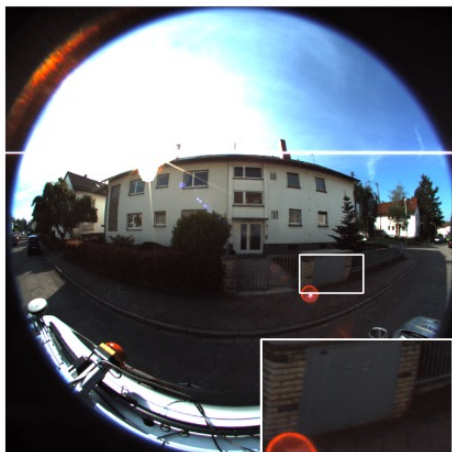
Ours

GT

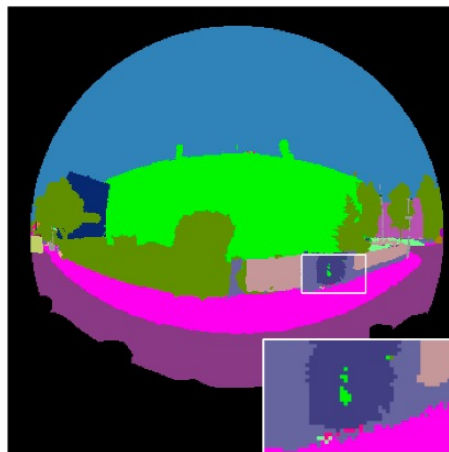
- It is more clear that 3D-2D CRF struggles in far and overexposed regions.

Neural Scene Representation

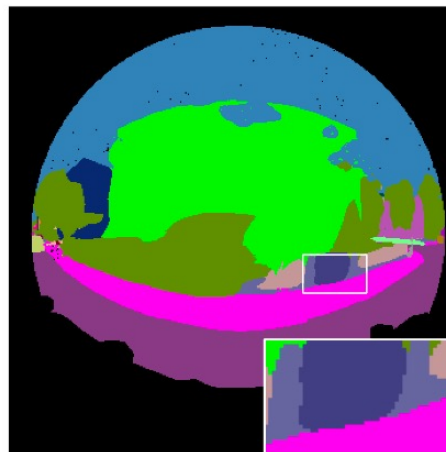
	Forward-facing			Fisheye		
Method	mIoU $\uparrow$	PQ $\uparrow$	PSNR $\uparrow$	mIoU $\uparrow$	PQ $\uparrow$	PSNR $\uparrow$
MLP [47]	81.2	69.5	23.78	68.2	55.8	24.47
iNGP [49]	79.0	67.9	27.22	67.0	54.6	28.14
Tri-planes [9]	75.1	62.1	23.25	60.2	50.2	23.32
Ours	<u>80.9</u>	<u>68.7</u>	27.25	<u>67.6</u>	<u>55.4</u>	<u>28.05</u>



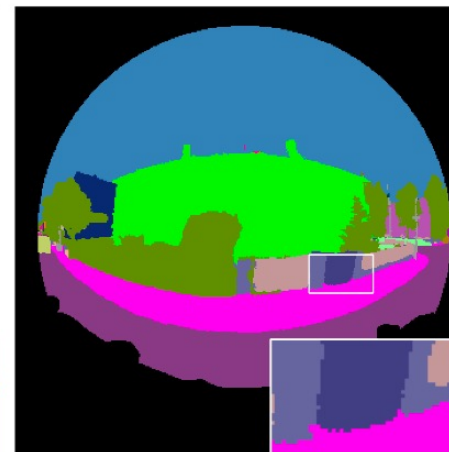
Input



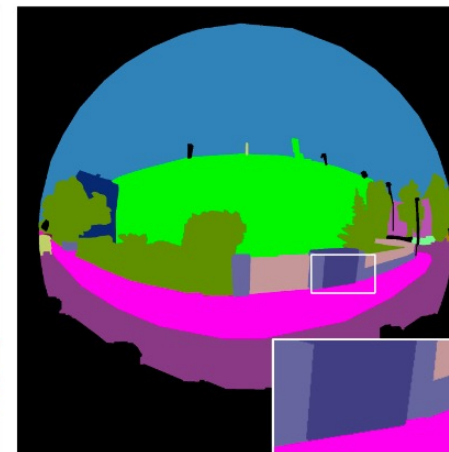
iNGP [49]



Tri-planes [9]



Ours



GT Label

Panoramic Label Synthesis



360° Outward Rotated Label & Appearance Synthesis

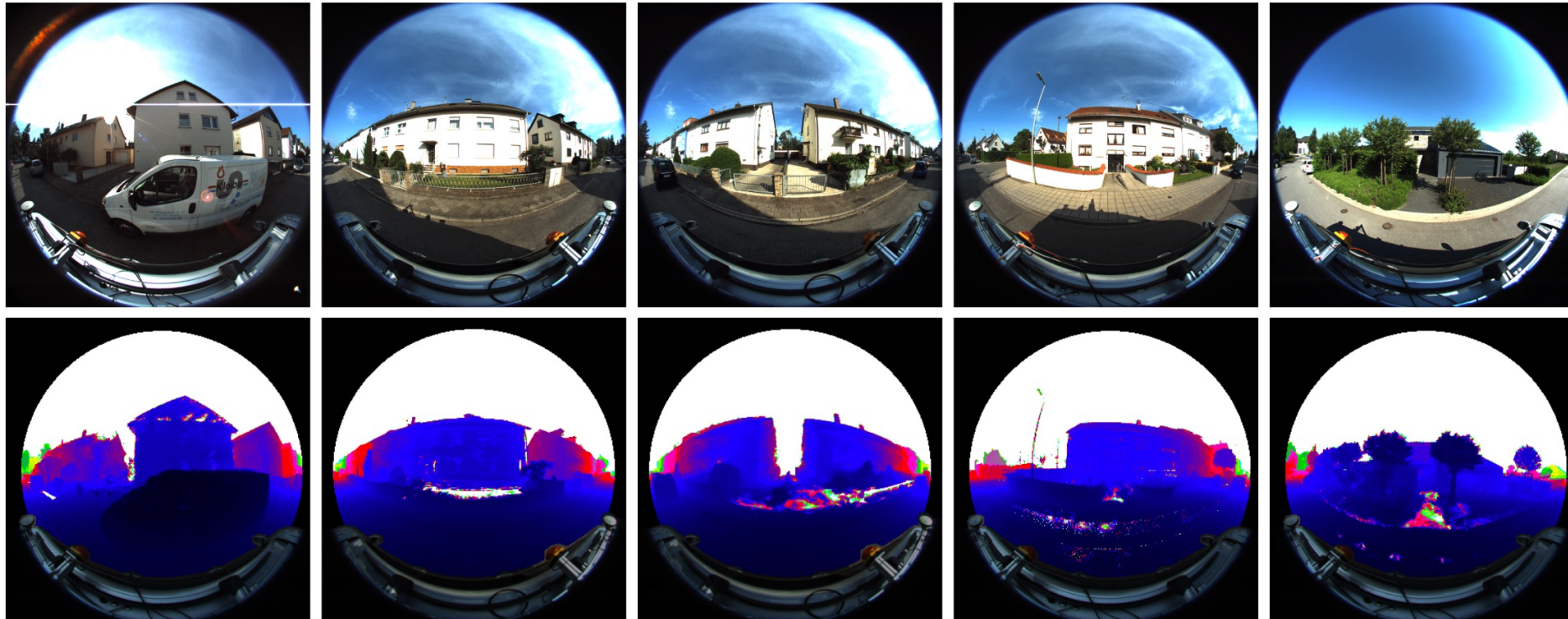


Failure cases

- Label synthesis in far-region at novel viewpoints.

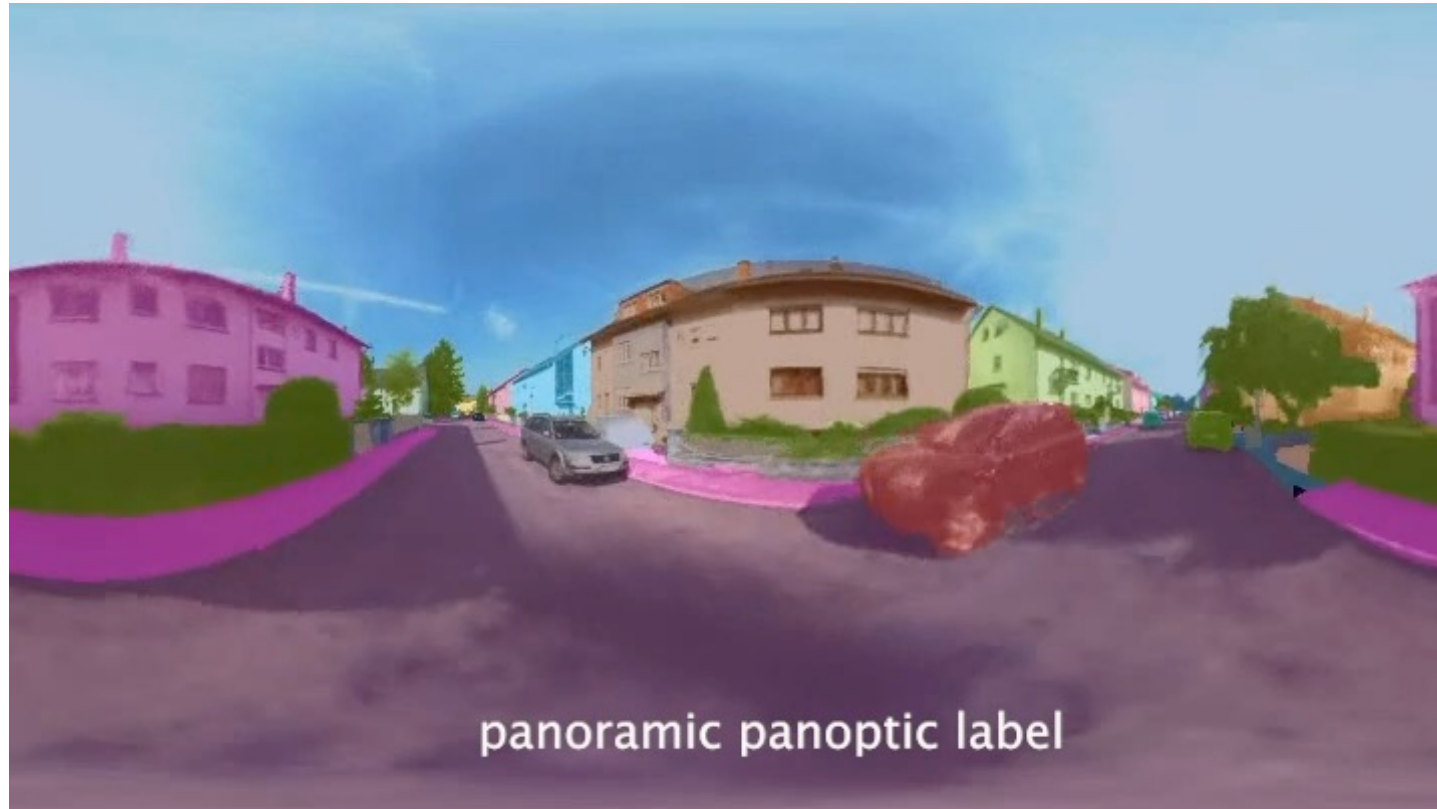


- Geometric reconstruction on fisheye views



Conclusion

- Efficient propagate coarse 3D annotations to 2D labels via neural rendering
- Enable omnidirectional rendering of high-fidelity, multi-view and spatiotemporally consistent appearance, semantic and instance labels
- propose to utilize mutual improvement of geometry and semantics



panoramic panoptic label

Future Work



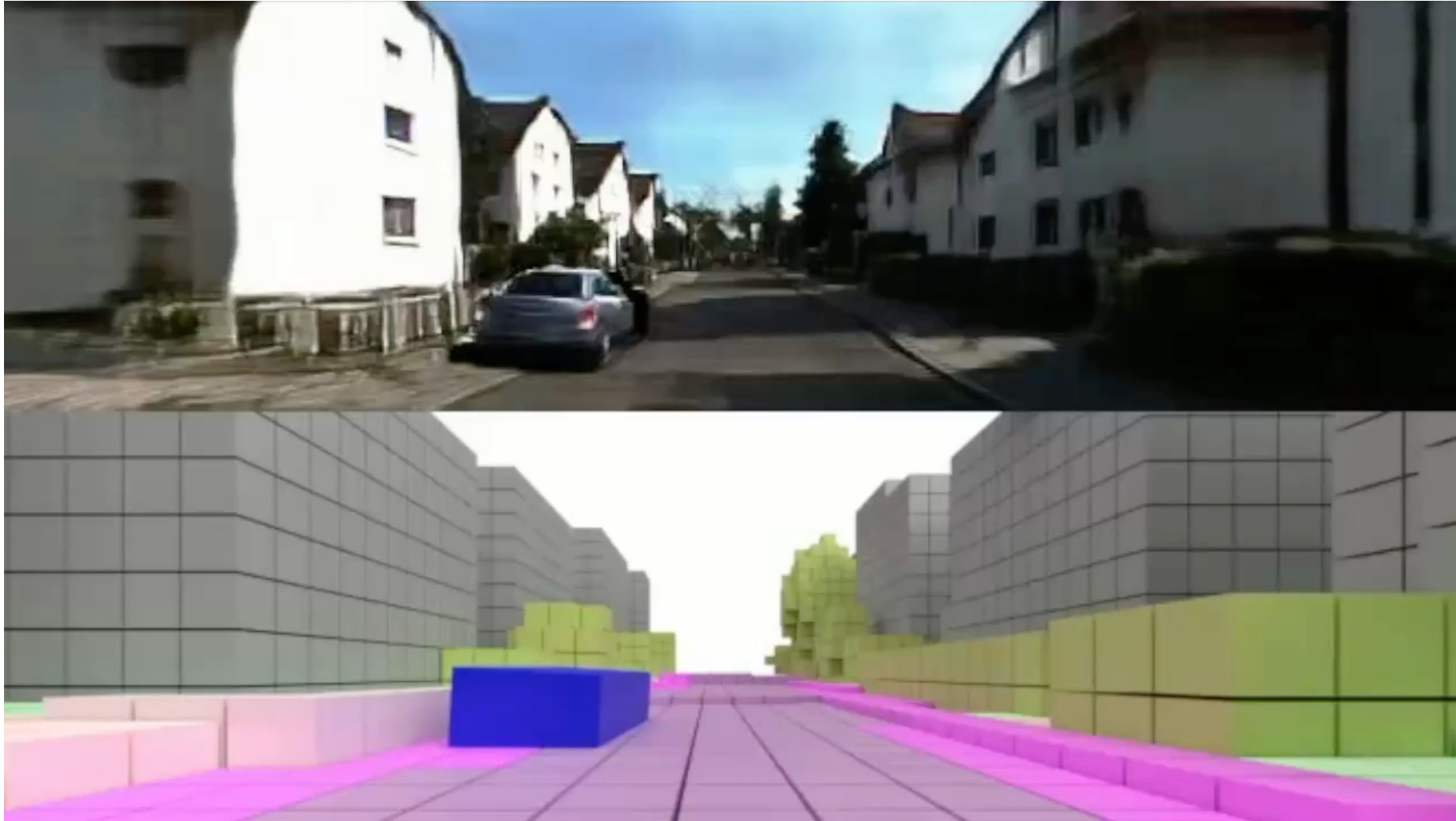
► Extra and store real-world assets, and further build more complex real-world-like scenarios

Future Work



► Dynamic 4D scene understanding and manipulation

Future Work



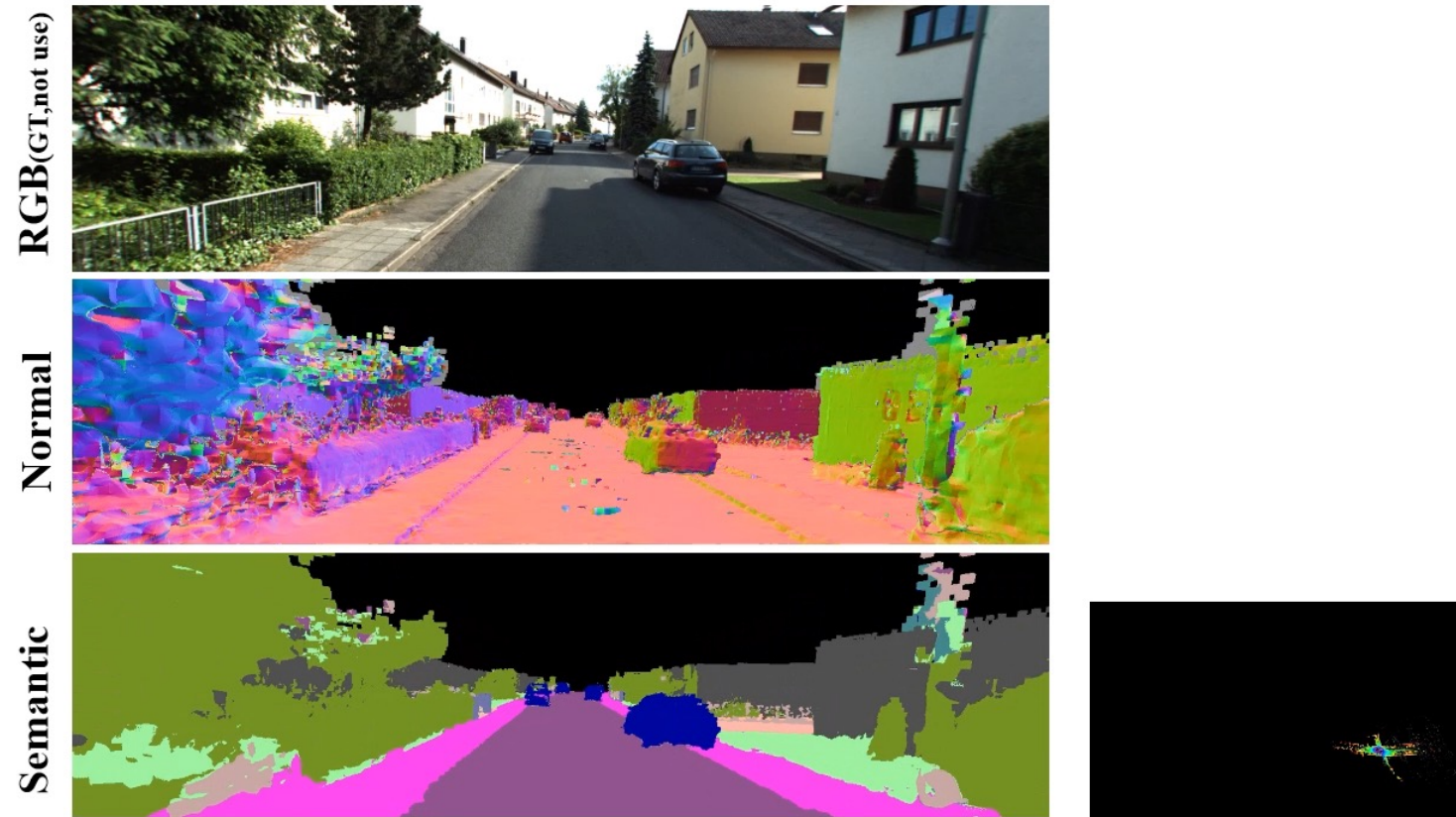
- Generate more assets on complex foreground & background

Future Work



► Integration of generated assets and real-world scenes

Future Work



► Semantic reconstruction at large-scale

Q & A

Thank you!

fuxiao0719.github.io/projects/panopticnerf360/